

A Reversible Smoothing Framework for Degenerate Riemannian Metrics

A Finite-Dimensional Analogue of Nash–Moser Theory via Lie Group Actions

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(Dates of server timestamps)*

Abstract

In this paper, “global optimum” or “global optimum’ ” is used as shorthand for "global optimum or a tolerance criterion was met" , so this means, for either (i) the true global minimizer or (ii) any point that satisfies the stopping tolerance criterion. In[1] we gave a link between the Nash Embedding Theorem and the barren plateaus problem of quantum computing. In this paper we give more evidence of this link. In this paper the metric maybe the quantum Fisher/Fubini information metric which is the metric associated with Born machines. We develop a finite-dimensional analogue of the Nash–Moser smoothing paradigm for the study of degenerate Riemannian metrics. Given a smooth parameter manifold M mapped into a Riemannian ambient space $(\mathcal{N}, h)$ via an embedding function $f_0 : M \rightarrow \mathcal{N}$, the induced pullback metric $g = f_0^* h$ may be rank-deficient on large regions (analogous to the “loss of derivatives” or “barren plateaus”).

We construct a family of reversible smoothing operators by perturbing the embedding along the flow of an ambient Lie group action:

$$f_\varepsilon(x) = \Phi_\varepsilon(f_0(x)) \quad (10)$$

whose perturbed pullback metrics

$$g_\varepsilon = f_\varepsilon^* h \quad (9)$$

become strictly positive-definite for sufficiently small $\varepsilon > 0$.

This produces a reversible metric inflation: a temporary embedding of (M, g) into the higher-dimensional ambient space $\mathcal{N}$, geometrically expanding the degenerate locus into a positively curved auxiliary topology (analogous to a bouquet of spheres), restoring non-degeneracy while preserving reversibility.

The construction is inspired by a reversible deformation mechanism originally arising in a quantum-algorithmic context. Here we extract and formalize the underlying geometric structure using Lie algebras, proving new results on:

- reversible extrinsic smoothing flows
- auxiliary metric inflation via commutator bounds
- a finite-dimensional “reverse Nash flow”

- We present the discovery of our Lie orbit bouquet complex (LOBC) which a new object previously remained undiscovered, in the Barren plateau landscapes or Barren plateau research, for example they LOBCs occur in Born machines by utilizing this LOBC we show that an infinite number of types of Born machines are trainable Where these Born machines have Barren Plateau's in their manifold. We do not avoid the barren plateaus because we address the problem directly of how to train a Born machines even if it's associated manifold contains barren plateaus. The results in this paper also applies to other VQAs , which have Barren Plateaus, apart from Born machines.
- Our paper introduces a second type of degenerate point that has no analogue in the original Nash setting. It is a conceptual extension, not a minor variation. The 1950s Nash paper(solving the isometric embedding problem for Riemannian manifolds) has one type; our paper has two, and the second type is near term more useful for physics.

The resulting framework establishes a rigorous, purely geometric dictionary for regularizing degenerate optimization landscapes.

1 On Foundational Priority

Trilogy level Foundational Priority Declaration. The three companion papers (i) A Reversible Smoothing Framework for Degenerate Riemannian Metrics, (ii) A Geometric Nash–Moser Implicit Function Theorem for PDEs, and (iii) Adaptive Architecture Quantum Meta-learning via Unitary Homotopy, collectively establish the complete architectural, geometric, analytic, and algorithmic foundations of the Reversible Nash–Moser / Unitary Homotopy paradigm.

Across this trilogy, we introduce the core mechanisms of the framework—including but not limited to reversible Lie-group smoothing flows, auxiliary metric inflation, bouquet-of-spheres topology, Lie-algebra smoothing, the Geometric Nash–Moser IFT, the Reversible Implicit Function Theorem (RIFT), the Quantum Nash–Moser Escape Principle, ITS-triggered reversible pSWAP deformations, macroscopic smoothing operators, and the reversible deform–solve–revert cycle—together with all derivative extensions of these mechanisms. This foundational claim explicitly encompasses the extension of these mechanisms to resolve landscapes containing multiple global minima and finite attractor sets, and applies universally to general parameterized quantum circuits, including all Variational Quantum Algorithms (VQAs) beyond standard Born machines.

These three papers jointly define the foundational structure of the reversible smoothing paradigm in geometry, PDE theory, and quantum variational optimization. Subsequent works may refine proofs, extend special cases, or introduce minor technical variations, but such developments are derivative elaborations of the unified framework established in this trilogy..

This paper establishes the complete architectural, algebraic, and geometric

framework for the Reversible Nash Flow, including but not limited to reversible extrinsic smoothing flows, auxiliary metric inflation via commutator bounds, the bouquet-of-spheres auxiliary topology, the reverse Nash flow, and all derivative extensions of these mechanisms.

While subsequent works may refine proofs, extend special cases, or introduce additional coordinate level formalizations, such developments are derivative elaborations of the framework fully defined in this manuscript. This paper therefore serves as the primary reference for the reversible smoothing architecture and its geometric foundations.

Introduction

Operator Correspondence and Scope. In this paper, “global optimum” is used as shorthand for either (i) the true global minimizer or (ii) any point that satisfies the stopping tolerance criterion. The algorithm does not attempt to distinguish between these cases. All smoothing operators used in this manuscript may be chosen to be exact Lie-group exponentials of the form $U(\varepsilon) = \exp(-i\varepsilon V)$, with V lying in a finite-dimensional Lie algebra. The reversible smoothing operators introduced in the companion quantum paper [1] are formally defined in Appendix E of that work, where the discrete pSWAP gate is replaced by its continuous $SO(2)$ Lie-group analogue $U_{\text{pSWAP}}(\varepsilon) = \exp(-i\varepsilon V_{\text{swap}})$. This construction provides an exact Lie-algebraic correspondence between the quantum smoothing operator and the geometric framework developed here, and therefore all convergence proofs in the present manuscript apply without modification. Other smooth Lie-group operators (e.g., shear-type or phase-scaling generators) could also be used within the same framework, but are not required for the results established below.

This work is inspired by a reversible deformation mechanism introduced in a quantum-algorithmic setting in [1]. A unitary perturbation was used to escape regions of vanishing curvature (“barren plateaus”) in a variational landscape. In this work/trilogy, we interpret barren plateaus as degenerate regions of the Quantum Fisher Information Metric (QFIM). This degeneracy arises in two complementary ways. First, real quantum hardware can only measure expectation values and gradients to finite precision. When the true gradient becomes exponentially small, it is overwhelmed by shot noise, readout noise, and gate noise, causing the measured gradient to register as zero. Operationally, the optimizer experiences a loss of directional derivatives, and the QFIM becomes effectively rank deficient. We assume the widely accepted conjecture that general quantum circuits cannot be efficiently simulated classically.

Second, for realistic classes of noise channels, such as depth dependent depolarizing or dephasing noise, the Born machine state can lose sensitivity to certain parameters. In these regimes, the QFIM becomes mathematically rank deficient, producing true degenerate points on the variational manifold. Therefore, the “degenerate point” interpretation is not merely a modeling convenience: it is rigorously valid for physically relevant noise models and empirically valid

whenever gradients fall below the hardware noise floor. These analytic and effective degeneracies constitute the finite dimensional analogue of derivative loss in Nash–Moser theory, motivating our reversible smoothing framework for escaping barren plateaus. Here we extract the underlying geometric mechanism and develop a fully mathematical formulation independent of quantum mechanics.

A point may also be regarded as degenerate in the limit when the baseline geometric term becomes progressively negligible under depth scaling, noise accumulation, or parameter growth, so that the Lie algebraic curvature injection term is the only contribution that remains dynamically significant. In such situations, the non-dominant term effectively fades away as the relevant physical or algorithmic parameters increase, while the dominant commutator term continues to supply curvature. When this imbalance persists under continued scaling, the point behaves as a limit type degenerate point even if no exact analytic degeneracy is present. This notion captures the physically important case where degeneracy emerges asymptotically rather than exactly, and it is treated identically to analytic degeneracy within our reversible smoothing framework.

We study smooth manifolds M mapped into an ambient space whose induced Riemannian metric g is degenerate on large subsets. Such degeneracies arise naturally in optimization, geometric statistics, and information geometry.

A fundamental restriction of differential geometry is that pulling back a degenerate metric by an intrinsic diffeomorphism on M cannot change its rank. To restore positive-definiteness, one must act extrinsically, inflating the manifold into a higher-dimensional ambient space. This mirrors John Nash’s use of ambient Euclidean spaces to resolve metric singularities.

Our goal: construct a reversible, extrinsic smoothing operator that temporarily inflates the degenerate metric into a strictly positive-definite form utilizing Lie group actions.

Main contributions:

- An extrinsic smoothing operator f_ε acting via an ambient Lie group.
- A reversible metric inflation theorem:

$$g_\varepsilon = f_\varepsilon^* h$$

becomes strictly positive-definite for small ε .

- A geometric interpretation of (M, g_ε) as a bouquet of hyperspheres.
- A reverse Nash flow resolving the “loss of derivatives” via algebraic tame estimates.

Preliminaries and Definitions

2.1 The Ambient Space and Lie Group Action

Let $(\mathcal{N}, h)$ be an N -dimensional Riemannian manifold (the ambient space). Let $\mathcal{G}$ be a Lie group acting isometrically on $\mathcal{N}$, and let $\mathfrak{g}$ be its Lie algebra.

For any $A \in \mathfrak{g}$ and $p \in \mathcal{N}$, the infinitesimal action generates a fundamental vector field on $\mathcal{N}$, denoted by $A \cdot p$.

2.2 The Parameter Manifold and Target Map

Let M be a smooth manifold of dimension d .

Let $\mathcal{U} : M \rightarrow \mathcal{G}$ be a smooth map associating each point in M to a group element.

Fix a base point $p_0 \in \mathcal{N}$.

The target embedding is the smooth map $f_0 : M \rightarrow \mathcal{N}$ defined by:

$$f_0(x) = \mathcal{U}(x) \cdot p_0 \quad (8)$$

2.3 The Degenerate Baseline Metric

The baseline metric on M is the pullback of the ambient metric:

$$g = f_0^* h \quad (1)$$

For $v \in T_x M$, the differential is $df_0(v) = d\mathcal{U}(v) \cdot p_0$.

Defining the local Lie algebra generator $A(v) = \mathcal{U}(x)^{-1} d\mathcal{U}(v) \in \mathfrak{g}$, the metric evaluates to:

$$g(v, v) = h(A(v) \cdot p_0, A(v) \cdot p_0) \quad (2)$$

The metric g is degenerate at x if there exists a non-zero $v \in T_x M$ such that $A(v) \cdot p_0 = 0$. The subset where this occurs is the degeneracy locus, $D \subseteq M$.

2.4 Tame Non-Commutativity Condition

To resolve the degeneracy, we select a global smoothing generator $V \in \mathfrak{g}$.

The Lie bracket in $\mathfrak{g}$ is $[V, A]$.

We say V satisfies the tame non-commutativity estimate if, for all v in the null space of the baseline metric ($A(v) \cdot p_0 = 0$), the commutator does not annihilate the base point:

$$\|[V, A(v)] \cdot p_0\|_h \geq c \|v\| \quad (3)$$

for some structural constant $c > 0$. This inequality guarantees the injection of curvature into the degenerate subspace.

Construction of the Extrinsic Smoothing Operator

We construct the smoothing operator not as a coordinate transformation on M , but as a continuous perturbation of the embedding into $\mathcal{N}$.

For $V \in \mathfrak{g}$ and intensity $\varepsilon > 0$, define the perturbed target map $f_\varepsilon : M \rightarrow \mathcal{N}$:

$$f_\varepsilon(x) = \mathcal{U}(x) \exp(\varepsilon V) \cdot p_0 \quad (4)$$

At $\varepsilon = 0$, $f_0(x)$ is recovered. The inverse flow $\varepsilon \rightarrow 0$ perfectly deflates the perturbation.

3.1 Pullback of the Inflated Metric

The perturbed metric on M is:

$$g_\varepsilon = f_\varepsilon^* h \quad (20)$$

For a tangent vector $v \in T_x M$, the perturbed differential is:

$$df_\varepsilon(v) = d\mathcal{U}(v) \exp(\varepsilon V) \cdot p_0 = \mathcal{U}(x) A(v) \exp(\varepsilon V) \cdot p_0 \quad (21)$$

Because $\mathcal{G}$ acts isometrically, the action of $\mathcal{U}(x)$ preserves the norm h . Thus, the quadratic form of the perturbed metric is:

$$g_\varepsilon(v, v) = h(A(v) \exp(\varepsilon V) \cdot p_0, A(v) \exp(\varepsilon V) \cdot p_0) \quad (22)$$

Auxiliary Metric Inflation

We now compute the metric expansion to prove inflation.

We rewrite the action on p_0 by inserting the identity $\exp(\varepsilon V) \exp(-\varepsilon V)$:

$$A(v) \exp(\varepsilon V) \cdot p_0 = \exp(\varepsilon V) (\exp(-\varepsilon V) A(v) \exp(\varepsilon V)) \cdot p_0 \quad (23)$$

Recognizing the Adjoint representation, $\text{Ad}_{\exp(-\varepsilon V)}(A(v)) = \exp(-\varepsilon V) A(v) \exp(\varepsilon V)$. Because $\exp(\varepsilon V)$ is an isometry, it can be dropped from the norm evaluation:

$$g_\varepsilon(v, v) = h(\text{Ad}_{\exp(-\varepsilon V)}(A(v)) \cdot p_0, \text{Ad}_{\exp(-\varepsilon V)}(A(v)) \cdot p_0) \quad (24)$$

4.1 The Taylor Expansion of the Adjoint

Expanding the Adjoint representation yields:

$$\text{Ad}_{\exp(-\varepsilon V)}(A(v)) = A(v) - \varepsilon[V, A(v)] + \mathcal{O}(\varepsilon^2) \quad (25)$$

Assume v is a degenerate direction, meaning $A(v) \cdot p_0 = 0$. The zero-order term vanishes entirely, isolating the commutator:

$$\text{Ad}_{\exp(-\varepsilon V)}(A(v)) \cdot p_0 = -\varepsilon[V, A(v)] \cdot p_0 + \mathcal{O}(\varepsilon^2) \quad (26)$$

Substituting this into the metric:

$$g_\varepsilon(v, v) = \varepsilon^2 \|[V, A(v)] \cdot p_0\|_h^2 + \mathcal{O}(\varepsilon^3) \quad (27)$$

By the tame condition, $\|[V, A(v)] \cdot p_0\|_h \geq c\|v\|$.

Thus:

$$g_\varepsilon(v, v) \geq c^2 \varepsilon^2 \|v\|^2 > 0 \quad (28)$$

The degenerate direction has been strictly inflated into a positively curved coordinate. A concrete way to compute ε is given in appendix E or F.

The Reverse Nash Flow

In classical NET, the iteration protocol introduces progressively high-frequency metric deformations to force an isometric embedding. Our framework executes an Inverse Nash Flow:

- $g_0 = g$ is pathologically degenerate.
- The flow f_ε acts as a non-local embedding functor, pushing the manifold out into the ambient Lie group symmetries.
- g_ε becomes strictly positive-definite, restoring differentiability and resolving the “loss of derivatives.”
- Linear reversion $\varepsilon \rightarrow 0$ systematically deflates the auxiliary scaffolding, crumbling the manifold back to its physical constraints.

Main Theorems

Theorem 6.1 Existence of the Positive-Definite Auxiliary Manifold

Let $g = f_0^* h$ be highly degenerate. If $V \in \mathfrak{g}$ satisfies the tame non-commutativity estimate $\|[V, A_i] \cdot p_0\|_h \geq c > 0$ for all local generators, then for a sufficiently small smoothing intensity $\varepsilon > 0$, the perturbed metric tensor g_ε of the auxiliary manifold is strictly positive-definite.

Theorem 6.2 Reversibility

The family of mappings f_ε satisfies $\lim_{\varepsilon \rightarrow 0} f_\varepsilon^* h = g$, ensuring that optimization over the auxiliary topology resolves deterministically back to the baseline constraints.

Theorem 6.3 Bouquet of Spheres Topology

For sufficiently small ε , the local geometry of (M, g_ε) near the degeneracy locus is diffeomorphic to a sub-manifold of $\mathcal{N}$ spanning the orthogonal commutator directions $[V, A_i] \cdot p_0$. Topologically, this acts as a bouquet of hyperspheres:

$$S^{(k_1)} \vee S^{(k_2)} \vee \dots \vee S^{(k_m)} \quad (29)$$

utilizing the higher-dimensional “ballooning” geometry purely as an optimization catalyst.

References

[1] Locate reference

Appendix A. Example: Resolving a Degenerate Metric on $SO(3)$

To visualize this geometrically, let $\mathcal{N} = \mathbb{R}^3$ with standard Euclidean metric h . Let $\mathcal{G} = SO(3)$. Let $p_0 = (0, 0, 1)^T$ (the North Pole).

Let $M = \mathbb{R}^2$ with coordinates (x, y) .

Define $\mathcal{U}(x, y) = \exp(xJ_z)\exp(yJ_z)$, where J_z generates rotations around the Z-axis.

The target map is:

$$f_0(x, y) = \exp((x + y)J_z) \cdot (0, 0, 1)^T = (0, 0, 1)^T \quad (\text{A1})$$

Because J_z stabilizes p_0 , the entire manifold M maps to a single degenerate point. The baseline metric is $g = 0$.

We choose the smoothing operator $V = J_x$ (rotation around X). The perturbed map is:

$$f_\varepsilon(x, y) = \exp((x + y)J_z) \exp(\varepsilon J_x) \cdot p_0 \quad (\text{A2})$$

The term $\exp(\varepsilon J_x) \cdot p_0$ tilts the pole away from the Z-axis by angle ε , creating a circle of radius $\sin(\varepsilon)$. The parameters (x, y) now actively trace this non-degenerate circle.

The metric expands exactly as predicted:

$$g_\varepsilon = \varepsilon^2(dx + dy)^2 + \mathcal{O}(\varepsilon^3) \quad (\text{A3})$$

The degenerate point has inflated into a well-conditioned geometric scaffold.

Appendix B. (Global reachability via Lie-orbit branching)

B1. Theorem B1 (Global reachability via Lie-orbit branching)

Note the theorems in all of this appendix B apply to local minima and barren plateaus or actually all known types of traps (e.g. degenerate metric regions or ill-conditioned saddles) because the stagnation detection is not dependant on the type of trap. In practice, escape from a nonglobal local minimum or a flattened barren plateau region occurs precisely when the ITS criterion detects stagnation, triggering a reversible inflation–deflation cycle that transports the state back to the degeneracy locus and into a new auxiliary balloon. Let (N, h) be a smooth Riemannian manifold, G a Lie group acting isometrically on N , and M a smooth parameter manifold with a smooth embedding

$$f_0 : M \rightarrow N, \quad g := f_0^*h \quad (\text{B1})$$

whose pullback metric g is allowed to be degenerate on a closed subset $\Sigma \subset M$ (the degeneracy locus). Let $V \in \mathfrak{g}$ be a fixed Lie algebra element and, for $\varepsilon \in \mathbb{R}$ small, define the perturbed embeddings

$$f_\varepsilon(x) := \Phi_\varepsilon(f_0(x)), \quad \Phi_\varepsilon := \exp(\varepsilon V) \in G, \quad (\text{B2})$$

with auxiliary metrics $g_\varepsilon := f_\varepsilon^* h$.

Let $F : M \rightarrow \mathbb{R}$ be a smooth cost function. Consider the piecewise-smooth dynamics obtained by alternating:

1. gradient flow of F with respect to g_ε on each auxiliary branch, and
2. reversible “inflation–deflation” steps $\varepsilon \mapsto 0$ that return the state to Σ when a plateau or non-global local minimum is detected.

Assumptions

(A1) Commutator tame estimate on the degeneracy locus: For every $x \in \Sigma$ and every nonzero $v \in T_x M$ with $g(v, v) = 0$, the associated generator $A(v) \in \mathfrak{g}$ satisfies

$$\|[V, A(v)] \cdot p_0\|_h \geq c\|v\| \quad (\text{B3})$$

for some constant $c > 0$ and fixed base point $p_0 \in N$. Equivalently, the auxiliary metric g_ε is strictly positive-definite on Σ for all sufficiently small $\varepsilon \neq 0$.

(A2) Orbit transversality at the degeneracy locus: For every $x \in \Sigma$, the derivatives of the family $\{f_\varepsilon\}$ span the ambient tangent space along the image of x :

$$\text{span}\{df_\varepsilon(x)[T_x M] \mid \varepsilon \in (-\varepsilon_0, \varepsilon_0)\} = T_{f_0(x)} N. \quad (\text{B4})$$

In particular, the Lie orbit through $f_0(x)$ is transversal to the degenerate directions of g .

(A3) Bouquet coverage of the search space: There exists a finite or countable family of auxiliary “balloons” $\{B_k\}_{k \in K} \subset M$ such that:

- each B_k is an open subset on which some g_{ε_k} is strictly positive-definite and the gradient flow of F is well-posed;
- each B_k intersects Σ (i.e., $\Sigma \cap B_k \neq \emptyset$);
- the union $\bigcup_{k \in K} B_k$ covers M (or at least the sublevel set $\{F \leq F(x_0)\}$ of interest).

(A4) Existence of at least one good auxiliary branch.

There exists a unique point $x^* \in M$ such that:

- x^* is the unique global minimizer of F ;
- there exists at least one auxiliary balloon B_{k^*} for which the negative gradient flow of F with respect to the corresponding auxiliary metric $g_{\varepsilon_{k^*}}$ has ω -limit set contained in the basin of attraction of x^* (with respect to the original metric structure);
- other balloons may contain nonglobal minima, saddles, plateaus, or degenerate regions, and therefore their gradient flows need not converge to x^* .

- **(A6) Patience-Calibrated Stagnation Detection.** The stagnation window is chosen so that ITS cannot fire before an ε -neighborhood of x^* has been visited on any branch reaching the basin of x^* (continuous-time analogue of (P4')).

Then, for any initial point $x_0 \in M$, the piecewise-smooth “inflate–flow–deflate” dynamics, with incumbent $\hat{x}(t) := \arg \min_{s \leq t} F(x(s))$, satisfies $\hat{x}(t) \rightarrow x^*$ as $t \rightarrow +\infty$, even if the raw trajectory $x(t)$ is repeatedly evicted from the basin of x^* after visiting it.

- **(A5) Episode level history buffer.** During any search episode, each balloon carries a binary flag indicating whether it has been visited. Once visited, it is not selected again in that episode. The bouquet is finite. At the start of each new episode, all flags reset to “unvisited.”

Proof

Step 1: Local metric inflation near Σ . By assumption (A1) and our existing metric-inflation theorem (Theorem 6.1 in this paper), for each $x \in \Sigma$ there exists $\varepsilon_0 > 0$ such that, for all $|\varepsilon| < \varepsilon_0$, the auxiliary metric g_ε is strictly positive-definite in a neighborhood of x . Thus, every degenerate direction of g at Σ is “inflated” into a non-degenerate direction for g_ε .

Step 2: Star-graph (bouquet) structure. By (A3), the family $\{B_k\}$ forms a collection of positively curved auxiliary regions, each intersecting Σ . Topologically, this realizes a “star-graph” or bouquet structure: Σ acts as a hub, and each B_k is a branch (or “balloon”) attached to it. The reversible inflation–deflation mechanism ensures that whenever a trajectory reaches Σ , it can be lifted into any of the intersecting B_k by choosing an appropriate ε_k .

Step 3: Escape from nonglobal critical points. Let x_{loc} be a strict local minimum of F that is not global. If the trajectory enters a neighborhood of x_{loc} inside some B_i , the gradient flow with respect to g_{ε_i} may converge to x_{loc} . By construction of the algorithm, once the “Iteration–Time–Stuck” (ITS) criterion detects stagnation (vanishing gradient or plateau), the reversible inflation–deflation step is triggered: the state is transported back along the Lie orbit to a point in Σ . Because the flow is reversible and norm-preserving at the level of the ambient action, this return does not erase the global structure of the landscape, only the local trapping. See appendix F for computing ε in a concrete way.

Step 4: Transversal branching from Σ . By (A2), at each $x \in \Sigma$ the derivatives of the embeddings f_ε span the ambient tangent space. In particular, the Lie orbit through $f_0(x)$ provides a transversal cone of directions that are not tangent to the degenerate subspace. Combined with (A1), this implies that for each visit to Σ there exists at least one auxiliary branch B_k such that the lifted point lies in a region where the gradient of F (with respect to g_{ε_k}) is nonzero and points away from previously explored basins of attraction. Thus, the algorithm can “branch” into a new balloon B_k that is dynamically distinct from the previously visited one.

Step 5: Coverage and eventual entry into the basin of x^* . By (A3), the union of balloons $\bigcup_k B_k$ covers the relevant portion of the search space. By (A4'), the global minimizer x^* is the unique global attractor, and there exists at least one auxiliary balloon B_{k^*} whose gradient flow converges to x^* . Because other balloons may contain nonglobal minima, saddles, plateaus, or degenerate regions, the trajectory may enter such balloons; however, each time ITS detects stagnation, the inflation–deflation step returns the trajectory to E , from which a new balloon is selected. Since the balloons cover the relevant sublevel set and the Lie orbit is transversal, repeated branching cannot avoid the good balloon B_{k^*} indefinitely. Thus, after finitely many cycles, the trajectory enters B_{k^*} at a point lying in the basin of attraction of x^* . Under (A5), each balloon in the finite bouquet can be visited at most once per episode. Since (A4) guarantees the existence of a good balloon, repeated branching cannot avoid indefinitely. Therefore the trajectory reaches the good balloon in at most branching steps.

Step 6: Asymptotic convergence. Once the trajectory lies in the basin of x^* for some auxiliary metric $g_{\varepsilon_{k^*}}$, the negative gradient flow of F with respect to $g_{\varepsilon_{k^*}}$ converges to x^* as $t \rightarrow +\infty$ by (A4). The reversibility of the smoothing flow ensures that subsequent deflation steps do not destroy this convergence: they only change the metric representation, not the location of the minimizer. Therefore, the ω -limit set of the full piecewise-smooth dynamics is $\{x^*\}$.

This proves global reachability of the unique global minimizer under assumptions (A1)–(A4). $\square$

Lemma 1 (Conditional Polynomial Lemma) *Assuming conditions (P1)–(P5), and assuming the ITS-based stagnation detection mechanism described in (P4), the search terminates in polynomial time. The polynomial bound is conditional on the tame-commutator regime established in Appendix E.*

Lemma 2 (Polynomial-Time Convergence of Theorem B1) *Assume the setting and hypotheses (A1)–(A4) of Theorem B1. Suppose, in addition, that the following quantitative conditions hold.*

(P1) Polynomial Bouquet Cardinality. *The auxiliary bouquet $\{B_k\}_{k \in K}$ contains*

$$m = |K| = \text{poly}(n),$$

where $n = \dim M$ (or the number of circuit parameters).

(P2i) Uniform Metric Equivalence (Dimension-Independent). *For each balloon B_k , the auxiliary metric g_k^ε satisfies*

$$\lambda_{\min} \|v\|^2 \leq g_k^\varepsilon(v, v) \leq \lambda_{\max} \|v\|^2,$$

with constants $\lambda_{\min}, \lambda_{\max} > 0$ independent of both k and n .

(P2ii) Uniform Metric Equivalence (Polynomial Dependence). *Alternatively,*

$$\lambda_{\min}(n)\|v\|^2 \leq g_k^\varepsilon(v, v) \leq \lambda_{\max}(n)\|v\|^2,$$

where $\lambda_{\min}(n)^{-1}$ and $\lambda_{\max}(n)$ grow at most polynomially in n . The strong-convexity constant $\mu(n)$ and smoothness constant $L(n)$ may also scale polynomially with n . The convergence rate depends on the condition number

$$\kappa(n) = \frac{L(n)}{\mu(n)},$$

which is assumed to satisfy $\kappa(n) = \text{poly}(n)$.

(P3) Gradient Regularity. *On each balloon B_k , the cost function F is L -smooth and μ -strongly convex with respect to g_k^ε .*

(P4) Controlled Branching and ITS Stagnation Detection. *Each time the inflation–deflation step returns the trajectory to the degeneracy locus E , the algorithm selects a new balloon B_k that has not previously been identified as leading to nonglobal convergence, unless the current point already lies in the basin of attraction of x^* . The ITS mechanism acts as a stagnation-detection oracle: under (P2)–(P3), ITS detects stagnation within a bounded number of steps and with a bounded false-negative rate. The polynomial-time guarantee is conditional on this stagnation-detection behavior.*

(P5) Polynomial Jump Cost

The reversible Lie-group smoothing operator

$$\Phi_\varepsilon : \mathcal{M} \rightarrow \mathcal{M}, \quad \Phi_\varepsilon(x) = f_\varepsilon(x),$$

and its inverse satisfy

$$\text{Time}(\Phi_\varepsilon), \text{Time}(\Phi_\varepsilon^{-1}) = \mathcal{O}(\text{poly}(n)).$$

Thus each inflation–deflation cycle has polynomial computational cost.

Conclusion

Under assumptions (A1)–(A4) and (P1)–(P5), and assuming either (P2i) or (P2ii), the inflate–flow–deflate dynamics reaches an ε -neighborhood of the global minimizer x^* in at most

$$T(\varepsilon) \leq \text{poly}(n, \lambda_{\min}^{-1}(n), \lambda_{\max}(n), \kappa(n)) + \mathcal{O}\left(\log \frac{1}{\varepsilon}\right).$$

The reversible Nash–Moser smoothing scheme therefore converges in polynomial time and uses only polynomial space; each balloon B_k is exited in at most $T_{\text{exit}} = \mathcal{O}(\text{poly}(n))$ steps by the ITS thresholding mechanism, which triggers a deflation–inflation cycle if the gradient norm $\|\nabla F\|$ remains below the stagnation threshold for a fixed duration

(P4') Patience-Calibrated Stagnation Detection (strengthens P4 for the purposes of the polynomial time lemma). The ITS stagnation duration D is chosen so that

$$D \geq C_1 \kappa(n) \log(1/\varepsilon)$$

for a fixed constant C_1 , i.e. at least the number of steps that (P2)–(P3) already guarantee suffice for ε -accuracy inside any single balloon. Consequently, if the trajectory enters the good balloon B_{k^*} , ITS cannot evict it before an ε -neighborhood of x^* has been visited.

Best-iterate reporting. The algorithm maintains the incumbent $\hat{x}_t := \arg \min_{s \leq t} F(x_s)$ over every point visited in every balloon, and outputs $\hat{x}_T$ in place of the final iterate x_T . Evaluating F costs nothing extra, since it is already computed at each gradient step.

Under assumption (P4'), the conclusion of the lemma takes the form

$$\min_{j \in X} \|x^T - x_j\| \leq \varepsilon,$$

rather than a statement about the final iterate itself. No other part of the argument above is impacted in the sense (P4') constrains only the timing of ITS activation, not the exploration dynamics of the algorithm.

(For A3 used in above lemma 2, in this appendix B, see footnote 1.)

Proof. [Proof Sketch] **Polynomially many balloons**

By (P1) and (P4), the trajectory can visit at most $m = \text{poly}(n)$ balloons before entering one whose basin contains x^* .

Geometric convergence inside a balloon

By (P2i)–(P3) or (P2ii)–(P3),

$$\|x_{t+1} - x^*\| \leq (1 - \eta\mu(n))\|x_t - x^*\|,$$

so reaching ε -accuracy requires $\mathcal{O}(\kappa(n) \log(1/\varepsilon))$ steps.

Polynomial jump cost

By (P5), each inflation-deflation step costs $\mathcal{O}(\text{poly}(n))$.

Combine

Total time = polynomially many balloons + $\mathcal{O}(\kappa(n) \log(1/\varepsilon))$ steps inside the correct balloon. Thus the runtime and memory usage are polynomial in n . ■

Note in this paper we sketch the proof a tame estimate in the section below (note-there is more than one way to prove this) in the geometric setting. Note, in our quantum companion paper we established a related or analogous probabilistic tame lemma showing that a random smoothing generator satisfies the required commutator bound almost surely .

B2. On the Tame Estimate

In our framework, we establish the tame estimate for the regularization operator by considering a randomly chosen generator V within our finite-dimensional

Lie algebra $\mathcal{G}$. Given that our Lie algebra possesses a trivial center, for any fixed non-zero generator A_i from our set $\{A_1, \dots, A_m\}$, we define the centralizer $\text{Cent}(A_i)$ as the subspace $\{V \in \mathcal{G} : [V, A_i] = 0\}$. Since $\text{Cent}(A_i)$ is a proper linear subspace of our finite-dimensional space, it holds a Lebesgue measure of zero. We observe that because our set of generators is finite, the union of these measure-zero subspaces is likewise of measure zero. Therefore, we conclude that for a randomly chosen V drawn from any absolutely continuous distribution, the probability that all commutators $[V, A_i]$ are non-zero is exactly 1.

Furthermore, we define our commutator norm as

$$X_i(V) := ||[V, A_i]||_F \quad (\text{B8})$$

Since each $X_i(V)$ is a continuous function, the global minimum

$$c(V) := \min_{1 \leq i \leq m} X_i(V) \quad (\text{B9})$$

remains a continuous, non-zero random variable almost surely. Consequently, we demonstrate that a randomly chosen V satisfies the required tame non-commutativity condition with probability 1. This result provides the rigorous foundation for our metric inflation, ensuring that the tame estimate holds for our smoothing operator and allowing us to resolve the derivative loss inherent in our degenerate geometric setting. We can construct an alternative proof that relies on the structural properties of our Lie algebra $\mathcal{G}$ —specifically through the Cartan-Weyl decomposition—rather than the measure-theoretic arguments we presented in this paper. While our probabilistic tame lemma guarantees that a random choice of V succeeds almost surely, the Cartan-Weyl approach allows us to define deterministic structural requirements on V that guarantee the tame estimate holds.

The almost sure nonzero commutator result is qualitative and does not imply an inverse polynomial lower bound. It supports Regime F (exponential), not Regime E (polynomial) in Appendix F and Appendix E.

B3. Multi-hub bouquet model for pSWAPs

In this part of the appendix we give complementary convergence arguments. The single-hub proof given earlier in the appendix serves as a "conceptual warm-up": it demonstrates polynomial convergence under a bouquet-of-spheres topology that can be extended as done here. The pSWAP architecture, however, generates a network of hubs, one for each reversible deformation. To capture this structure faithfully, we now introduce a multi-hub convergence proof. The new proof contains the earlier one as a special case and shows that the same polynomial-time guarantees hold in the full Unitary Homotopy setting.

.Let:

- Parameter space: M (e.g. circuit parameters).

- Cost function: $F : M \rightarrow \mathbb{R}$, smooth.
- Global minimizer: $x^* \in M$, unique.

We model the pSWAP-based architecture as follows.

Hubs and balloons

Hubs: there are a finite family of “degeneracy loci”

$$E(j) \subset M, \quad j = 1, \dots, H, \quad (\text{B100})$$

each corresponding to a qualitatively distinct pSWAP configuration (or cluster of configurations).

Bouquets: for each hub $E(j)$, a family of open sets

$$\{B_k(j)\}_{k \in K_j} \subset M \quad (\text{B101})$$

(the “balloons”) such that:

- $E(j) \cap B_k(j) \neq \emptyset$ for all k ,
- on each $B_k(j)$ we have a pSWAP-inflated metric $g_k(j)$ (coming from the corresponding unitary deformation).

Total bouquet size:

$$m := \sum_{j=1}^H |K_j| = \text{poly}(n), \quad (\text{B102})$$

where n is the problem size (e.g. number of parameters / qubits).

Algorithmic dynamics (inflate–flow–deflate with pSWAPs)

The algorithm evolves as a piecewise process:

Flow phase (inside a balloon):

If $x_t \in B_k(j)$, we run (discrete) gradient descent on F with respect to the metric $g_k(j)$:

$$x_{t+1} = \Phi_k(j)(x_t), \quad (\text{B103})$$

where $\Phi_k(j)$ is one GD step (or a few steps) under $g_k(j)$.

Stagnation detection (ITS):

If the ITS criterion triggers (gradient norm small, no progress in value, etc.), we deflate back to the hub:

$$x_t \mapsto \pi(j)(x_t) \in E(j), \quad (\text{B104})$$

via the inverse pSWAP deformation (the “reversible” step).

Branching to a new balloon:

From $E(j)$, we choose a new balloon $B_{k'}(j')$ (possibly with a different hub j') and inflate via a new pSWAP deformation:

$$x_t \mapsto x_{t+1} \in B_{k'}(j'). \quad (\text{B105})$$

This is the “inflate–flow–deflate” process, for multiple hubs and pSWAP-induced bouquets.

2. Assumptions (multi-hub analogue of A1–A4, P1–P5)

We now impose assumptions that mirror above proof, but for the multi-hub pSWAP setting.

(MH1) Metric inflation (tame non-commutativity). For every hub $E(j)$, every balloon $B_k(j)$, and every $x \in E(j) \cap B_k(j)$, the pSWAP-inflated metric $g_k(j)$ is strictly positive-definite on all directions that are degenerate under the baseline metric.

(MH2) Orbit transversality. At each hub $E(j)$, the derivatives of the pSWAP-induced embeddings span enough directions so that the union of balloons attached to $E(j)$ is transversal to the degenerate directions. Intuitively: each time we return to a hub, there exist balloons that point us out of the plateau.

(MH3) Bouquet coverage. The union of all balloons across all hubs covers the relevant sublevel set:

$$\bigcup_{j,k} B_k(j) \supset \{x \in M : F(x) \leq F(x_0)\}, \quad (\text{B106})$$

for the initial point x_0 .

(MH4) Existence of at least one good balloon per hub. There is a unique global minimizer x^* , and for each hub $E(j)$ there exists at least one balloon $B_{k^*}(j)$ such that the gradient flow of F with respect to $g_{k^*}(j)$ has its ω -limit set contained in the basin of attraction of x^* (with respect to the baseline geometry). Other balloons attached to the same hub may contain nonglobal minima, saddles, plateaus, or degenerate regions, and therefore their gradient flows need not converge to x^* .

(MH5) Polynomial bouquet cardinality. As above, $m = \sum_j |K_j| = \text{poly}(n)$.

(MH6) Controlled branching. Each time the algorithm returns to a hub $E(j)$, if the current point is not already in the basin of x^* , it eventually selects a new balloon $B_{k'}(j')$ that has not been visited before (or at least not revisited indefinitely). So we cannot get stuck cycling among a small set of bad balloons forever.

(MH7) Nice geometry inside each balloon. On each balloon $B_k(j)$, the function F is: $L(n)$ -smooth and $\mu(n)$ -strongly convex with respect to $g_k(j)$, with condition number $\kappa(n) = L(n)/\mu(n) = \text{poly}(n)$, and the metric $g_k(j)$ is uniformly equivalent to the Euclidean metric with polynomially bounded constants.

(MH8) Polynomial cost per cycle. Each inflate–flow–deflate cycle (including pSWAP application and its inverse) has computational cost $\text{poly}(n)$.

These above are tied to pSWAP-induced geometry.

Note operationally, from this section two it follows the role of a tabu-style memory: the algorithm avoids revisiting previously explored balloons that failed to produce descent.

3. Theorem (Polynomial-time convergence for multi-hub pSWAP bouquet model)

(MH6') Patience-Calibrated Controlled Branching. Strengthens (MH6): the ITS window at each hub satisfies $D \geq C_1 \kappa(n) \log(1/\varepsilon)$, so the trajectory cannot be evicted from a good balloon before visiting an ε -accurate point. The algorithm maintains the incumbent $\hat{x}_t := \arg \min_{s \leq t} F(x_s)$ across all hubs and balloons.

Theorem. Assume (MH1)–(MH5), (MH6'), (MH7)–(MH8). Then the multi-hub inflate–flow–deflate dynamics, reporting $\hat{x}_T$, satisfies $\|\hat{x}_T - x^*\| \leq \varepsilon$ in at most $T(\varepsilon) \leq \text{poly}(n, \kappa(n)) + O(\log(1/\varepsilon))$ steps, robust to further ITS activations after x^* 's balloon is visited

4. Proof (adapting the above proof to multiple hubs)

Step 1: Bound the number of distinct balloons visited.

By (MH5), the total number of balloons is

$$m = \sum_{j=1}^H |K_j| = \text{poly}(n). \quad (\text{B108})$$

By (MH6) (controlled branching), each time the algorithm returns to a hub and is not yet in the basin of x^* , it eventually selects a new balloon. Therefore, the algorithm can visit at most m distinct balloons before either:

- entering the basin of attraction of x^* , or
- exhausting all balloons (which is ruled out by (MH3) and (MH4), see below).

So the number of inflate–flow–deflate cycles before entering a “good” balloon whose flow leads to x^* is at most $m = \text{poly}(n)$.

Step 2: Existence of at least one “good” balloon.

By (MH3) (coverage), the union of all balloons covers the relevant sublevel set $\{F \leq F(x_0)\}$. By (MH4’), for each hub $E(j)$ there exists at least one good balloon $B_{k^*}(j)$ whose gradient flow converges to x^* . Other balloons attached to the same hub may contain nonglobal minima or degenerate regions. Because controlled branching (MH6) prevents indefinite revisiting of bad balloons, the algorithm must eventually enter a good balloon at some hub., there exists at least one balloon $B_{k^*}(j^*)$ such that:

- $B_{k^*}(j^*)$ intersects the basin of attraction of x^* ,
- and the gradient flow of F with respect to $g_{k^*}(j^*)$ from any point in that intersection converges to x^* .

Thus, among the m balloons, there is at least one “good” balloon whose flow leads to x^* .

By (MH6), the algorithm cannot avoid this balloon forever: after at most m visits to hubs and corresponding branchings, it must enter $B_{k^*}(j^*)$ at a point in the basin of x^* .

Step 3: Geometric convergence inside the good balloon.

Once the algorithm is in the good balloon $B_{k^*}(j^*)$ at a point in the basin of x^* , we consider the subsequent gradient descent steps with respect to $g_{k^*}(j^*)$. By (MH7), F is $\mu(n)$ -strongly convex and $L(n)$ -smooth with respect to $g_{k^*}(j^*)$, with condition number $\kappa(n) = L(n)/\mu(n) = \text{poly}(n)$. Standard results for gradient descent on strongly convex functions then give:

$$\|x_{t+1} - x^*\| \leq (1 - c/\kappa(n))\|x_t - x^*\| \quad (\text{B109})$$

for some constant $c > 0$ and suitable step size. Hence, to reach an ε -neighborhood of x^* , we need

$$O(\kappa(n) \log(1/\varepsilon)) \quad (\text{B110})$$

gradient steps inside this balloon. See appendix E for computing ε in a concrete way.

Step 4: Total number of steps and total cost.

- Number of cycles before entering the good balloon: at most $m = \text{poly}(n)$.
- Number of gradient steps inside the good balloon: $O(\kappa(n) \log(1/\varepsilon)) = \text{poly}(n) \log(1/\varepsilon)$.
- Cost per cycle (including pSWAP inflate/deflate): $\text{poly}(n)$ by (MH8).

Therefore, the total number of steps and the total computational cost are bounded by

$$T(\varepsilon) \leq \text{poly}(n) + O(\kappa(n) \log(1/\varepsilon)) \quad (\text{B111})$$

with each step costing $\text{poly}(n)$, so overall polynomial in n and logarithmic in $1/\varepsilon$.

Note the tame non-commutativity estimate proved above for the pSWAP generator extends to each hub in the multi-hub bouquet model. Accordingly, Assumption MH1 in the expanded convergence theorem is satisfied, and the polynomial-time result follows

This is exactly the multi-hub analogue of the polynomial-time convergence lemma above, now explicitly expressed for a model where each pSWAP gate (or sequence) is allowed to define its own hub with bouquet structure. **End of proof.**

We can add the following, using **MH9** and **MH10**, to the above proof for transitioning between hubs.

(MH9) History-buffer-guided branching:

The algorithm maintains a finite history buffer $H \subset \{(j, k)\}$ of hub-balloon pairs previously identified as non-productive. Whenever ITS triggers at hub $E(j)$, the next balloon $B_{k'}(j')$ is chosen such that $(j', k') \notin H$, ensuring non-redundant exploration across hubs and balloons.

(MH10) Reachability via reversible pSWAP homotopies:

For any hub $E(j)$ and any balloon $B_{k'}(j')$ with $E(j') \cap B_{k'}(j') \neq \emptyset$, there exists a finite sequence of reversible pSWAP-based unitary homotopies that transports the state from $E(j)$ into $B_{k'}(j')$ in polynomial time.

The assumptions (MH9) and (MH10) extend the multi-hub framework by incorporating the history-buffer-guided branching mechanism and an explicit reachability condition between hubs. When these assumptions are used, the algorithm explores the hub-balloon structure in a non-redundant manner and can transition between distinct hubs via reversible pSWAP-based homotopies. Under these extended assumptions, the polynomial-time convergence guarantees of the multi-hub model remain valid.

Our global convergence results apply not only to barren plateaus modeled as QFIM degeneracies, but also to search spaces containing traditional analytically flat degenerate regions, true “tabletop type” surfaces where the gradient is exactly zero but which do not satisfy the formal expectation and variance definition of barren plateaus. In such mixed landscapes, the reversible Lie group smoothing injects curvature into the flat degenerate regions, while the ITS triggered branching mechanism ensures escape from (McClean) type barren plateaus. Consequently, even when the search space contains a combination of McClean/formally defined barren plateaus and analytically flat degenerate points, the inflate–flow–deflate dynamics remains unobstructed and still guarantees eventual reachability of the global optimum (implies a guarantee of trainability). One can construct explicit examples containing both types of traps, along with a global minimizer, to illustrate and verify that the convergence theorem holds in all such scenarios. Note that this proof requires access to a Born Machine and may be applied to a search landscape for which the geometry

is not known before (so if the search landscape satisfies the assumptions the polynomial time proof of global convergence here will apply).

We can optionally consider the following remark 1 which convert the above Theorem (Polynomial-time convergence for multi-hub pSWAP bouquet model) into a probabilistic version of the proof.

Remark 1. The preceding proof establishes a strict polynomial-time bound under the assumption of deterministic, tabu-guided descent. However, this framework can be naturally generalized into a probabilistic walk by introducing a hub-dependent jump probability, $\alpha_w \in (0, 1]$, for each hub w . In this stochastic variant, the algorithm obeys the history-buffer-guided descent with probability $1 - \alpha_w$, but with probability α_w , it abandons the local descent gradient to initiate a random jump to an unexplored state. This mechanism dynamically constructs a probabilistic network-or “random graph”-of explored trajectories, which is highly advantageous for escaping local minima in non-convex landscapes. However, introducing this stochasticity fundamentally alters the theoretical time complexity. Because the algorithm is no longer strictly bound to a monotonically decreasing path, the random walk can theoretically wander indefinitely before finding the global minimum. Consequently, in the general case, any non-zero α_w transforms the strict polynomial-time guarantee into an expected exponential-time bound. Polynomial-time convergence within this stochastic regime can only be recovered under highly restrictive conditions, such as implementing a rapid decay factor where $\alpha_w \rightarrow 0$ fast enough to strictly bound the maximum allowable number of probabilistic jumps to a polynomial constant.

Remark 5. Scope of the polynomial result. The polynomial-time convergence theorem in this Appendix B is not claimed as a generic resolution of the barren plateau problem. It applies only to search landscapes (or circuit families) that are explicitly constructed to satisfy structural assumptions (P1)–(P5) and the tame commutator regime of Appendix E. These assumptions define an constructed, controlled class of landscapes for which our reversible smoothing framework yields polynomial-time convergence. For generic deep circuits exhibiting barren plateaus, Appendix F shows that exponential-time escape guarantees are available.

Footnote 1. A closely related coverage condition appears in the classical proof of the Nash Embedding Theorem for compact manifolds. Compactness guarantees that one may extract a finite subcover from the coordinate atlas, and Nash’s local corrugation steps are applied on patches whose supports collectively cover the entire manifold. Assumption (A3) plays the analogous role here: the auxiliary balloons must cover the effective search region (e.g., the relevant sublevel set). Without such coverage, the inflate–flow–deflate dynamics could enter regions where no auxiliary metric is available, and the global reachability and polynomial-time convergence arguments would fail.

Remark 6

The Natural Emergence of the Star-of-Stars Structure. A remark on the nature of the assumptions underlying our convergence theorems is warranted here, because it resolves what might otherwise appear as a tension in the framework. The search landscapes to which our theorems apply are described as “constructed” in the sense that assumptions (A1)–(A5) and (P1)–(P5) are conditions one imposes, rather than properties that hold universally for every parameterized quantum circuit. One might therefore conjecture that the geometric structure our framework uncovers – the hub-and-spoke bouquet, the star-of-stars branching, the balloon topology – is itself an artefact of those assumptions, something we built in by hand rather than something genuinely discovered. This conjecture is misplaced. The assumptions are the price of admission for *seeing* the structure clearly; they are not the source of the structure itself. What we found, and what we believe is the deeper contribution of this work, is that a hidden geometric architecture was already present in the Born machine and VQA landscape, waiting to be uncovered. It did not need to be constructed – it needed to be recognized. Three features of the mathematics make this precise. First, the star shape at each hub is not imposed: it is the natural geometry of the Lie group orbit fanning out from a degeneracy point in all generator directions simultaneously, a consequence of orbit transversality (A2) alone. Second, the star-of-stars structure of the multi-hub model is not a separate construction but the same single-star geometry applied recursively, once for each pSWAP configuration that defines its own degeneracy locus – the hierarchy emerges from iterating one idea, not from introducing a new one. Third, the bouquet-of-spheres topology of Theorem 6.3 is not assumed anywhere: it is *derived* from the commutator directions $[V, A_i] \cdot p_0$ spanning orthogonal subspaces of the ambient tangent space, with the spheres appearing as the level sets of the inflated metric in those directions. In this sense the artificial landscapes of our theorems are artificial only in the sense that we chose to look at them. The geometric structure we find inside them – the star orbits, the bouquet topology, the curvature injection from non-commutativity – was latent in the Lie group action all along, a hidden architecture lurking beneath the apparently featureless surface of the barren plateau. Our assumptions do not create this structure; they create the conditions under which it becomes visible. What the framework offers, at its core, is not a new construction imposed on the landscape from outside, but a lens through which a structure that was always intrinsically there can finally be seen and exploited. We refer to this structure formally as **Lie-orbit bouquet complex**. In formally it is a “bouquet constellation”.

We can optionally consider the following remark 1 which convert the above Theorem (Polynomial-time convergence for multi-hub pSWAP bouquet model) into a probabilistic version of the proof.

Remark 7a

Separation of the Three Geometric Levels and Exactness of the Quantum Correspondence. A potential source of confusion in reading the quantum companion paper [1] alongside the present work is the appearance of three

distinct spaces that must be kept carefully separate: the parameter manifold M , the ambient Riemannian manifold N , and the smoothing intensity parameter ε . We clarify each in turn and explain why their separation is essential for the exactness of the correspondence between the two papers.

Level 1: The parameter manifold M . The parameter manifold has fixed dimension d throughout the entire inflate-flow-deflate procedure. In the quantum setting, $M = \mathbb{R}^d$ with coordinates $\phi = (\phi_1, \dots, \phi_d)$, the variational parameters of the Born machine or variational quantum algorithm. The pSWAP deformation does not add new coordinates to M : the dimension of the parameter space being optimised is d before, during, and after every inflation-deflation cycle.

Level 2: The ambient space N . In the present geometric framework, N is a fixed Riemannian manifold on which the Lie group G acts isometrically. In the quantum setting, N is the projective Hilbert space $\mathcal{P}(H)$, which has fixed real dimension $2^{n+1} - 2$ for an n -qubit system. This space is fixed and does not change dimension at any point in the algorithm. The Born machine embedding $f_0(\phi) = U(\phi)|0\rangle$ maps M into $\mathcal{P}(H)$ as a d -dimensional submanifold. The pSWAP deformation $f_\varepsilon(\phi) = U(\phi)e^{i\varepsilon V_{\text{swap}}}|0\rangle$ maps M into a *different* d -dimensional submanifold of the *same* fixed $\mathcal{P}(H)$: it accesses directions of $\mathcal{P}(H)$ that were always present but unreachable under f_0 , without enlarging $\mathcal{P}(H)$ in any way. The auxiliary balloons and the bouquet-of-spheres topology constructed in Theorem 6.3 live entirely within this fixed ambient $\mathcal{P}(H)$: they are submanifolds of N , not extensions of it.

Level 3: The smoothing intensity ε and scheduling parameters. The smoothing intensity $\varepsilon > 0$ is an external algorithm control parameter - chosen by the ITS mechanism according to condition (P4') - and is not a coordinate of M nor a dimension of N . In the present paper it appears as the parameter of the one-parameter family of Lie group elements $\exp(\varepsilon V) \in G$, exactly as the smoothing radius appears in classical Nash-Moser theory. In the quantum companion paper [1], the macroscopic smoothing circuit

$$U_{\text{macro}}(\varepsilon) = \bigotimes_{i=1}^{w/2} I_1^{p_i} \cdot U_{\text{pSWAP}}(\varepsilon)^{q_i}$$

is a concrete quantum circuit realisation of the abstract geometric operator $\exp(\varepsilon V)$, under the exact Lie-algebra identification $V_{\text{geom}} \leftrightarrow iV_{\text{swap}}$ established in Appendix E.5 of [1]. The discrete scheduling parameters $p_i \in \mathbb{Z}$ and $q_i \in \{0, 1\}$ control the depth and activation of local pSWAP blocks within the circuit; they are implementation details of how the quantum hardware realises $\exp(\varepsilon V)$ and do not appear as separate geometric objects in the present proof, because the proof requires only the action of $\exp(\varepsilon V)$ on N - specifically its first-order Taylor expansion $\text{Ad}_{\exp(\varepsilon V)}(A(v)) = A(v) - \varepsilon[V, A(v)] + O(\varepsilon^2)$ and the resulting commutator bound (the tame estimate, Eq. (3)) - not the internal circuit structure that implements it. The parameters p_i and q_i are therefore absorbed into the definition of V_{swap} at the geometric level, and the correspondence remains exact. **So explaining, in part, why the correspondence is**

exact. The three levels above are related as follows. The parameter manifold M (Level 1) maps into the ambient space $N = \mathcal{P}(H)$ (Level 2) via the embedding f_ε , which is controlled by the smoothing intensity ε (Level 3). The pullback metric $g_\varepsilon = f_\varepsilon^* h$ is the exact QFIM of the Born machine initialised at the rotated base point $e^{i\varepsilon V_{\text{swap}}} |0\rangle$: not an approximation, not a model, but the literal Riemannian metric induced on M by the Fubini-Study geometry of $\mathcal{P}(H)$ via f_ε . When $\varepsilon = 0$, $f_0(M)$ is the original Born machine surface in $\mathcal{P}(H)$, possibly degenerate (a barren plateau). When $\varepsilon \neq 0$, $f_\varepsilon(M)$ is a different, non-degenerate surface in the same $\mathcal{P}(H)$: the auxiliary balloon. Deflation ($\varepsilon \rightarrow 0$) restores $f_0(M)$ exactly, with no residual smoothing and no permanent change to M , N , or the circuit architecture. This three-level separation is the geometric explanation, in part, the phrase “exact correspondence” used in Appendix E.5 of the quantum companion paper [1]. It is the quantum analogue of the isometric embedding condition $f^* h = g$ in Nash’s theorem: the extrinsic geometry of $\mathcal{P}(H)$ faithfully and completely represents the intrinsic optimisation geometry of M via the pullback, at every value of ε , exactly. No approximation enters at any level, and no dimension of M or $\mathcal{P}(H)$ changes at any point. The convergence theorems of Appendix B therefore apply to the quantum algorithm of [1] without modification, as stated in Appendix E.6 of that work.

Remark 7b Ambient Dimension Sufficiency

Remark 1 *The auxiliary balloon submanifold $f_\varepsilon(M)$ constructed by the pSWAP deformation lives within the same ambient projective Hilbert space $\mathcal{P}(H)$ as the original Born machine surface $f_0(M)$, without requiring any enlargement of H or $\mathcal{P}(H)$. This follows directly from the fact that the pSWAP operator $e^{i\varepsilon V_{\text{swap}}}$ is a unitary acting on the existing n -qubit Hilbert space H : it maps H to H by definition, so $e^{i\varepsilon V_{\text{swap}}} |0\rangle \in H$ and $f_\varepsilon(\phi) = U(\phi) e^{i\varepsilon V_{\text{swap}}} |0\rangle \in \mathcal{P}(H)$ for all ϕ and all ε . No independent assumption about the size of $\mathcal{P}(H)$ is required: the ambient space is automatically large enough to contain both $f_0(M)$ and $f_\varepsilon(M)$ simultaneously, because both are images of M under unitaries acting on the same fixed H . The real dimension $2^{n+1} - 2$ of $\mathcal{P}(H)$ exceeds the dimension d of M by an exponential factor for any $n \geq 2$, providing vastly more room than the inflated surface requires. This is the quantum analogue of Nash’s observation that a sufficiently high-dimensional Euclidean space always has room to accommodate an isometric embedding of any compact Riemannian manifold: here the ambient space $\mathcal{P}(H)$ is fixed by the physics of the n -qubit system and is automatically of sufficient dimension.*

APPENDIX C

Application of the Reversible Nash–Moser / Lie-Group Smoothing Framework to General Variational Quantum Algorithms

This appendix extends the geometric framework to variational quantum algorithms (VQAs), including VQE and QAOA.

C.1 VQAs as Degenerate Riemannian Manifolds

Any VQA defines a parameter manifold M with coordinates θ . The quantum state:

$$|\psi(\theta)\rangle = U(\theta) |0\rangle \quad (\text{C200})$$

induces the pullback metric:

$$g_{ij}(\theta) = \text{Re} \langle \partial_i \psi | \partial_j \psi \rangle, \quad (\text{C201})$$

the Quantum Fisher Information Metric (QFIM).

Barren plateaus correspond to:

$$g_{ij}(\theta) \approx 0, \text{ or } g_{ij}(\theta) = 0 \quad (\text{C202})$$

i.e., loss of directional derivatives, the finite-dimensional analogue of Nash–Moser derivative loss.

C.2 Explicit Non-Commutativity Requirement

Let $A(v)$ be the Lie-algebra generator associated with direction $v \in T_\theta M$. Let $V \in \mathfrak{g}$ be the smoothing generator. To inflate degenerate metric directions, we require:

$$[V, A(v)] \neq 0, \quad (\text{C203})$$

and the tame estimate:

$$\|[V, A(v)] \cdot p_0\| \geq c > 0. \quad (\text{C204})$$

This ensures:

$$g_\epsilon(v, v) = \epsilon^2 \|[V, A(v)] \cdot p_0\|^2 + \mathcal{O}(\epsilon^3) > 0. \quad (\text{C205})$$

Thus the QFIM becomes strictly positive-definite under the reversible deformation.

C.3 Specialization to VQE and QAOA

VQE

Hardware-efficient ansätze often have large commutative subalgebras. Choosing V outside these centralizers ensures curvature injection.

QAOA

QAOA’s alternating unitaries generate structured Lie algebras with symmetry-induced degeneracies. Choosing V such that:

$$[V, B] \neq 0, \quad [V, C] \neq 0, \quad (\text{C206})$$

ensures metric inflation.

C.4 Coherence Restoration

The inverse deformation:

$$U(-\epsilon) \tag{C207}$$

annihilates the smoothing operator:

$$U(-\epsilon)U(\epsilon) = I, \tag{C208}$$

restoring the original metric g and the original circuit architecture. This is essential for QAE, which requires a static, fully coherent unitary.

C.5 Summary

The geometric Nash–Moser framework applies directly to VQAs because:

- VQAs induce a Riemannian metric (QFIM)
- barren plateaus correspond to metric degeneracy
- non-commuting Lie-group smoothing inflates the metric
- the deformation is reversible
- coherence is restored by the inverse flow

Thus VQE and QAOA are natural specializations of the general reversible smoothing paradigm.

2 Appendix D. Extension to Multiple Global Minima

In Appendix B, the global reachability and polynomial-time convergence of the inflate-flow-deflate dynamics were established under Assumption (A4), which requires a unique global minimizer x^* . In many practical variational landscapes, particularly those arising in quantum optimization, the cost function exhibits symmetries that produce multiple, degenerate global minima.

This appendix extends the convergence guarantees to address search spaces containing a finite set of global minimizers. The structural mechanisms of the Reversible Nash Flow, including the bouquet-of-spheres topology and controlled branching, naturally accommodate this generalization without requiring additional computational overhead. The following theorem below first considers with using Remark 1 then after the theorem is given considers it.

2.1 D1. Theorem (Polynomial-Time Convergence with Multiple Global Minima)

Theorem 3 *Let M be a smooth parameter manifold, $F : M \rightarrow \mathbb{R}$ a smooth cost function, and let*

$$\mathcal{X}^* = \{x_1^*, x_2^*, \dots, x_m^*\}$$

denote the (finite) set of global minimizers of F . Assume hypotheses (A1)–(A3), (P1)–(P3), (P4'), (P5) of the corrected Theorem B1/Lemma 2, with the following modification of (A4)::

(A4') *(Finite global attractor set) There exists a finite set $\mathcal{X}^* \subset M$ of global minimizers such that, for every auxiliary metric g_k on each balloon B_k , the negative gradient flow of F with respect to g_k has its ω -limit set contained in the basin of attraction of at least one point $x_j^* \in \mathcal{X}^*$.*

Furthermore, assume that:

- *Each $x_j^* \in \mathcal{X}^*$ lies in the basin of attraction of at least one balloon B_k under the corresponding inflated metric g_k .*
- *The reversible inflation-deflation mechanism can reach at least one such balloon from any initial point $x_0 \in M$ via the hub structure and bouquet coverage.*

Then, under assumptions (A1)–(A3), (A4'), and (P1)–(P5), the inflate-flow-deflate dynamics converges to the global minimizer set $\mathcal{X}^$ in polynomial time and polynomial space. In particular, for any $\varepsilon > 0$, there exists a polynomial*

$$T(\varepsilon, n) = \text{poly}(n, \kappa(n), \log(1/\varepsilon))$$

such that

with incumbent $\hat{x}_t := \arg \min_{s \leq t} F(x_s)$, after at most $T(\varepsilon, n)$ iterations the algorithm outputs $\hat{x}_T$ satisfying $\min_{x_j^ \in \mathcal{X}^*} \|\hat{x}_T - x_j^*\| \leq \varepsilon$,*

where $n = \dim M$ and $\kappa(n)$ denotes the (polynomially bounded) condition number associated with the auxiliary metrics g_k .

If we use Remark 1 in the above Theorem given in appendix B then it follows we consider the following **Remark 2**.

Remark 2 on Stochastic Generalization under Multiple Optima While the deterministic tabu-search framework guarantees polynomial-time convergence to a valid global optimum within the multi-hub architecture, extending this to a probabilistic walk (via hub-dependent jump probabilities α_w) introduces distinct behaviors in landscapes with multiple global optima. In this multi-optima regime, a non-zero α_w no longer merely risks indefinite wandering; it permits the algorithm to dynamically transition between the basins of attraction of different equivalent global optima. While this stochastic "random graph" enhances global exploration and multi-modal discovery, it forfeits the strict deterministic upper bound on convergence time. Consequently, just as in the single-optimum case, any unconstrained stochasticity breaks the strict polynomial-time guarantee,

transforming the search into an expected exponential-time process across the broader set of multiple target states unless bounded by a rapid decay schedule ($\alpha_w \rightarrow 0$).

2.2 D2. Proof Sketch for Theorem D1

The proof follows the structural logic established in Appendix B (without using remark 1 in any way) for the single-attractor case, modified to account for the disjunction of target basins:

- **Polynomial Balloon Exhaustion:** By (P1) and the controlled branching assumption (P4), the algorithm visits at most a polynomial number of dynamically distinct balloons B_k before finding a target basin.
- **Coverage of Target Basins:** By the modified Assumption (A4') and the bouquet coverage properties, the finite set of target basins is accessible. The trajectory cannot infinitely cycle through non-global critical points because the transverse branching mechanism eventually routes the state into a balloon B_k whose gradient flow leads to at least one global minimizer $x_j^* \in \mathcal{X}^*$.
- **Local Geometric Convergence:** Once the algorithm enters a "good" balloon containing the basin for some x_j^* , the local landscape is $\mu(n)$ -strongly convex and $L(n)$ -smooth with respect to the inflated metric g_k . The gradient descent phase guarantees geometric convergence to that specific minimizer x_j^* , requiring $\mathcal{O}(\kappa(n) \log(1/\varepsilon))$ steps to reach the ε -neighborhood.
- **Total Complexity:** Summing the polynomial number of inflation-deflation jumps (each costing $\mathcal{O}(\text{poly}(n))$) and the geometric convergence steps inside the final balloon yields the stated polynomial total time $T(\varepsilon, n)$. End of (Sketch proof).

If we did use remark 1 above then we would have a stochastic version of the above proof which is described as follows in remark 3.

Remark 3 (Scope and Limitations on Stochastic Extensions): The proof sketch for Theorem D.1 relies strictly on the deterministic invariants of the multi-hub architecture—specifically, bounded bouquet cardinalities, history-buffer pruning, and controlled state-space traversal. These mechanisms guarantee that the search space is systematically filtered without cycles, yielding a strict polynomial-time bound to valid global optima across multiple basins of attraction.

Extending this formal proof sketch to accommodate the stochastic variant, wherein a hub-dependent jump probability $\alpha_w > 0$ introduces probabilistic walks and dynamic random graph topologies, fundamentally alters the mathematical nature of the analysis. In a purely deterministic regime,

path traversal is strictly bound by local geometry and memory-guided exclusion. Under stochastic rules, however, the system transitions from a bounded combinatorial path problem into a random process governed by state transitions.

Consequently, analyzing convergence with unconstrained α_w values across multiple optima requires shifting mathematical tools entirely: from worst-case combinatorial path bounds to Markov chain transition matrices, stationary distributions, and expected hitting or mixing times. Because such a formulation must account for the complex probabilities of escaping or re-entering various basins of attraction, it inevitably surrenders the strict deterministic polynomial-time guarantee in favor of an expected exponential-time bound, unless tightly constrained by a vanishing decay schedule ($\alpha_w \rightarrow 0$). Therefore, the formal guarantees established in Theorem D.1 apply exclusively to the deterministic baseline, leaving the rigorous treatment of the general stochastic multi-optima regime as a separate analytical extension.

Appendix E. Polynomial Lower Bound For Curvature

Regime Classification. Appendix E describes a special structural regime (Regime E) in which the commutator lower bound satisfies , yielding polynomial time convergence. Appendix F describes the generic regime (Regime F), where commutators may be exponentially small, yielding only exponential time guarantees. Our polynomial theorem applies only in Regime E; the exponential theorem applies in Regime F. This means we are not claiming polynomial behavior generically.

Assumption (Quantitative Non-Commutativity). Let $\mathfrak{g}$ be the Dynamical Lie Algebra (DLA) associated with a given family of parameterized quantum circuits. Assume that there exists a function

$$c_{\min}(n) \geq 1/\text{poly}(n) \tag{E320}$$

such that for all normalized generators $A \in \mathfrak{k}$ and $V \in \mathfrak{p}$ appearing in a Cartan decomposition $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$, and satisfying $[V, A] \neq 0$, the commutator obeys the quantitative lower bound

$$\|[V, A]\| \geq c_{\min}(n). \tag{E300}$$

This assumption defines the class of circuit families for which the reversible Nash–Moser / Unitary Homotopy smoothing mechanism yields polynomial-time escape from degenerate regions. A specific concrete circuit may be checked for the theorem here,

Theorem (Polynomial Lower Bound on Cartan-Injected Curvature).

Let $\mathfrak{g}$ be the DLA of a parameterized quantum circuit family acting on n qubits.

Assume:

- **Polynomial Dimension:** $\dim(\mathfrak{g}) = O(\text{poly}(n))$.
- **Cartan Decomposition:** $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ with

$$[\mathfrak{k}, \mathfrak{k}] \subseteq \mathfrak{k}, \quad [\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}, \quad [\mathfrak{k}, \mathfrak{p}] \subseteq \mathfrak{p}. \quad (\text{E301})$$

- **Generator Alignment:** The degenerate direction aligns with a normalized generator $A \in \mathfrak{k}$, $\|A\| = 1$, and the smoothing operator is a normalized element $V \in \mathfrak{p}$, $\|V\| = 1$.
- **Quantitative Non-Commutativity Structural Assumption** (this is an assumption about the chosen circuit architecture, not a derived theorem; the polynomial-time argument is conditional on this structural property). As stated above. To complete the polynomial-time argument, we record the following conclusion. Under the quantitative non-commutativity assumption stated above, the commutator lower bound satisfies

$$c_{\min}(n) \geq \frac{1}{\text{poly}(n)}, \quad (\text{E302})$$

for all directions relevant to the smoothing flow. This establishes a polynomial lower bound on the Cartan-injected curvature and ensures that the metric-inflation mechanism of Theorem 6.1 operates in the polynomial regime. Furthermore, the theorem applies to any Born machine ansatz family satisfying these structural conditions whose dynamical Lie algebras have polynomial dimension, admit a Cartan decomposition, and satisfy the tame-estimate hypotheses required for curvature inflation. For such families, the auxiliary balloons constructed in Appendix B cover the relevant sublevel sets, and the reversible smoothing flow preserves strong convexity within each balloon. Consequently, the polynomial-curvature bound above combines with the balloon-coverage assumptions to guarantee that the reversible deform–solve–revert cycle enters a good balloon in polynomially many steps, completing the polynomial-time convergence argument. Actually we need to use this with a specific point lets refine the above to express it with p_0 for clearness (a similar p_0 step is required for appendix F though not given but follows in a related way).

Refined Assumption (Quantitative Non-Commutativity).

- Assume there exists $c_{\min}(n) \geq 1/\text{poly}(n)$ such that, for every normalized degenerate generator $A(v) \in \mathfrak{k}$ and smoothing generator $V \in \mathfrak{p}$ with $[V, A(v)] \neq 0$,

$$\|[V, A(v)] \cdot p_0\|_h \geq c_{\min}(n) \|v\|, \quad (\text{E303})$$

i.e. the bound is on the commutator's action on the base point p_0 (the quantity in the tame estimate, Eq. (3)), not the abstract operator norm $\|[V, A]\|$.

Refined Conclusion.

- Under this assumption, $c_{\min}(n) \geq 1/\text{poly}(n)$ is the tame estimate of Theorem 6.1 with $c = c_{\min}(n)$, so the metric-inflation mechanism operates in the polynomial regime by direct substitution. (The operator-norm bound $\|[V, A]\| \geq c_{\min}(n)$ may be kept as a remark: necessary, but not sufficient on its own.)

Appendix F. Exponential Lower Bound For Curvature

Regime Classification. Appendix E describes a special structural regime (Regime E) in which the commutator lower bound satisfies , yielding polynomial time convergence. Appendix F describes the generic regime (Regime F), where commutators may be exponentially small, yielding only exponential time guarantees. Our polynomial theorem applies only in Regime E; the exponential theorem applies in Regime F. This means we are not claiming polynomial behavior generically.

- **Theorem (Exponential Lower Bound on Cartan-Injected Curvature).** Let $\mathfrak{g}$ be the Dynamical Lie Algebra (DLA) of a parameterized quantum circuit family acting on n qubits. Assume:

- **Polynomial Dimension:** $\dim(\mathfrak{g}) = O(\text{poly}(n))$.
- **Cartan Decomposition:** $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$ with

$$[\mathfrak{k}, \mathfrak{k}] \subseteq \mathfrak{k}, \quad [\mathfrak{p}, \mathfrak{p}] \subseteq \mathfrak{k}, \quad [\mathfrak{k}, \mathfrak{p}] \subseteq \mathfrak{p}. \quad (\text{E304})$$

- **Generator Alignment:** The degenerate direction aligns with a normalized generator $A \in \mathfrak{k}$, $\|A\| = 1$, and the smoothing operator is a normalized element $V \in \mathfrak{p}$, $\|V\| = 1$.
- **This is an assumption about the chosen circuit architecture, not a derived theorem; the exponential-time argument is conditional on this structural property. Assume there exists**

$$c_{\text{exp}}(\mathbf{n}) = \Omega(2^{-n}) \quad (\text{E305})$$

such that, for every normalized degenerate generator $A(v) \in \mathfrak{k}$ and smoothing generator $V \in \mathfrak{p}$ with $[V, A(v)] \neq 0$,

$$\|[V, A(v)] \cdot p_0\|_h \geq c_{\text{exp}}(n) \|v\|, \quad (\text{E306})$$

i.e., as in the Appendix E, the bound is on the commutator's action on p_0 (the quantity in the tame estimate, Eq. (3)), not the abstract operator norm $\| [V, A] \|$. Under this assumption, $c_{\text{exp}}(n) = \Omega(2^{-n})$ is the tame estimate of Theorem 6.1 with $c = c_{\text{exp}}(n)$, so by Eq. (28),

$$g_\varepsilon(v, v) \geq c_{\text{exp}}(n)^2 \varepsilon^2 \|v\|^2 > 0. \quad (\text{E307})$$

In particular: the smoothing step cannot produce curvature smaller than $O(2^{-n})$; the algorithm still injects strictly positive curvature in every degenerate direction; and global escape is guaranteed *under the patience-calibrated ITS condition below P4''*, with worst-case escape time exponential in n . (Note the operator-norm bound

$$\| [V, A] \| \geq c_{\text{exp}}(n) \quad (\text{E325})$$

is a remark is necessary and is not sufficient alone.)

This theorem is the weaker, more general counterpart of the polynomial lower-bound theorem: it uses the same Cartan-based mechanism, but with an exponential lower bound that is easier for a broad class of circuits to satisfy.

(P4'') Patience Calibration for Regime F.

Because injected curvature is only $c_{\text{exp}}(n) = \Omega(2^{-n})$, Theorem 6.1's bound forces the metric-equivalence constant of (P2i)/(P2ii) to inherit the same scaling, $\mu_{\text{exp}}(n) = \Theta(c_{\text{exp}}(n)^2) = \Theta(4^{-n})$, giving an exponential condition number

$$\kappa_{\text{exp}}(n) = \frac{L(n)}{\mu_{\text{exp}}(n)} = \Omega(4^n) \quad (\text{E308})$$

in place of the polynomial $\kappa(n)$ of Regime E. The patience window is recalibrated accordingly:

$$D_{\text{exp}} \geq C_1 \kappa_{\text{exp}}(n) \log(1/\varepsilon). \quad (\text{E309})$$

The algorithm maintains the same incumbent

$$\hat{x}_t := \arg \min_{s \leq t} F(x_s) \quad (\text{E310})$$

already defined for (P4'); with D_{exp} as above, ITS cannot evict the trajectory from a balloon containing a global minimizer before an ε -accurate point has been visited and locked into $\hat{x}$ – exactly as in the polynomial case, just with a longer patience window.

Conclusion

- As in the polynomial case, this escape guarantee requires the branching mechanism to avoid evicting the trajectory from a balloon containing a global minimizer before it has produced an ε -accurate point; this is ensured by the patience condition (P4'') below, with incumbent tracking as in (P4').

Corollary (Exponential-Time Convergence, Regime F).

Assume (A1)–(A5), (P1), (P3), (P4''), (P5), and the Revised Assumption above in place of (P2i)/(P2ii). Then, reporting the incumbent $\hat{x}_T$,

$$\min_{x_j^* \in X^*} \|\hat{x}_T - x_j^*\| \leq \varepsilon \quad (\text{E311})$$

after at most

$$T_{\text{exp}}(\varepsilon) \leq \text{poly}(n) \cdot 2^{O(n)} \cdot \log(1/\varepsilon) \quad (\text{E312})$$

total steps: $m = \text{poly}(n)$ balloons visited by (P1)/(P5), each exited in $O(\text{poly}(n))$ steps if bad, or run up to $O(\kappa_{\text{exp}}(n) \log(1/\varepsilon))$ steps if good. The procedure runs this fixed, precomputed budget and stops it never searches adaptively or detects completion at runtime. This is the Regime F counterpart of the Lemma 2: identical mechanism, exponential constants..

Appendix H. Validity Window for ε , Dominance of Injected Curvature, and Comparison with the McClean Barren Plateau Threshold

This appendix H adds more expository detail to the proof. The curvature injection argument requires ε to satisfy two simultaneous conditions, and the injected curvature must exceed the exponentially small threshold set by the McClean barren plateau definition.

(i) Upper bound (Taylor remainder dominance). The expansion

$$g_\varepsilon(v, v) = \varepsilon^2 \| [V, A(v)] \cdot p_0 \|_h^2 + O(\varepsilon^3)$$

is valid and the ε^2 term dominates the remainder only for

$$\varepsilon < \frac{c(n)^2}{C},$$

where $C > 0$ bounds the $O(\varepsilon^3)$ remainder. This is what “sufficiently small $\varepsilon > 0$ ” means throughout the proof.

(ii) Lower bound (dominance over the degenerate baseline and the McClean threshold). The McClean barren plateau condition (McClean et al., 2018) states that for a random parameterized quantum circuit on n qubits, the diagonal QFIM entries satisfy

$$g_{kk}(\phi) \leq \frac{C_0}{2^n}$$

in the plateau region, for a circuit-dependent constant $C_0 > 0$. In the approximately-zero case, the degenerate baseline metric satisfies $g(v, v) = \delta^2 \|v\|^2$ where $\delta^2 \leq C_0/2^n$. For the injected curvature to strictly dominate both the degenerate baseline and the McClean threshold, one requires

$$\varepsilon^2 c(n)^2 > \frac{C_0}{2^n},$$

i.e.

$$\varepsilon > \varepsilon_*(n) := \frac{\sqrt{C_0}}{c(n) \cdot 2^{n/2}}.$$

(iii) The validity window. Both conditions (i) and (ii) are simultaneously satisfiable: the window

$$\varepsilon_*(n) = \frac{\sqrt{C_0}}{c(n) \cdot 2^{n/2}} < \varepsilon < \frac{c(n)^2}{C}$$

is non-empty whenever

$$\frac{\sqrt{C_0}}{c(n) \cdot 2^{n/2}} < \frac{c(n)^2}{C},$$

which simplifies to $\delta \ll c(n)^3/C$, holding whenever the noise floor δ is small relative to the tame constant $c(n)$ – precisely the physical regime of a barren plateau.

(iv) Regime-dependent behavior of $\varepsilon_*(n)$.

- **Regime E** ($c(n) \geq 1/\text{poly}(n)$, say $c(n) = 1/n^r$):

$$\varepsilon_*(n) = \frac{\sqrt{C_0} \cdot n^r}{2^{n/2}} \rightarrow 0 \quad \text{exponentially fast.}$$

Any fixed $\varepsilon > 0$, however small, eventually exceeds $\varepsilon_*(n)$ for all n beyond a computable threshold n_0 . The injected curvature therefore dominates the McClean barren plateau threshold for all $n \geq n_0$.

- **Regime F** ($c(n) = \Omega(2^{-n})$, say $c(n) = c_0/2^n$):

$$\varepsilon_*(n) = \frac{\sqrt{C_0}}{c_0} = O(1),$$

a constant independent of n . The smoothing intensity ε must exceed this constant threshold, consistent with the exponential patience window of Appendix F. In the exactly-zero case ($\delta = 0$), the lower bound vanishes and any $\varepsilon \in (0, c(n)^2/C)$ suffices.

(v) Gradient recovery above the hardware noise floor. By the quantum Cramér–Rao bound,

$$\left| \frac{\partial F_\varepsilon}{\partial \phi_k} \right|^2 \leq \text{Var}[H] \cdot g_{\varepsilon, kk}.$$

Since $g_{\varepsilon, kk} \geq \varepsilon^2 c(n)^2 > C_0/2^n$ for all $\varepsilon > \varepsilon_*(n)$, the gradient of the inflated cost function F_ε is strictly larger than the McClean barren plateau bound, making it detectable above the hardware shot-noise floor for any K measurements satisfying $K > \text{Var}[H] \cdot 2^n/C_0$.

(vi) Summary. For any ε in the validity window $(\varepsilon_*(n), c(n)^2/C)$, the tame estimate (A1) gives

$$g_\varepsilon(v, v) \geq \varepsilon^2 c(n)^2 \|v\|^2 > \frac{C_0}{2^n} \|v\|^2 > 0$$

strictly, establishing that the injected curvature simultaneously: (a) is strictly positive in every degenerate direction; (b) dominates the approximately-zero degenerate baseline; and (c) exceeds the McClean barren plateau threshold $C_0/2^n$, making the gradient detectable above the hardware noise floor. The threshold $\varepsilon_*(n)$ is the precise quantitative condition under which the reversible Nash–Moser curvature injection overcomes the McClean barren plateau, and it is computable from the tame constant $c(n)$ and the circuit-dependent constant C_0 .

Appendix I. More on Connection To Nash Embedding Theorem/Isometric Embedding of Riemannian Manifolds

Exactness of the Geometric Correspondence.

- The auxiliary balloon geometry introduced by the pSWAP deformation is not an approximation to the Born machine's search landscape: it *is* the Born machine's search landscape, exactly, viewed from a rotated base point within the same ambient projective Hilbert space $\mathcal{P}(H)$. The deformed embedding $f_\varepsilon(\phi) = U(\phi) e^{i\varepsilon V_{\text{swap}}} |0\rangle$ defines an exact submanifold of $\mathcal{P}(H)$, with geometry inherited exactly from the Fubini-Study metric h via the pullback $g_\varepsilon = f_\varepsilon^* h$. The inflated metric g_ε is therefore not a regularisation or approximation of the QFIM: it is the exact QFIM of the Born machine initialised at the rotated state $e^{i\varepsilon V_{\text{swap}}} |0\rangle$.
- Our paper introduces a second type of degenerate point that has no analogue in the original Nash setting. It is a conceptual extension, not a minor variation. The 1950s Nash paper has one type; our paper has two, and the second type is more useful for near term physics.

Side by side comparison below.

- Nash: Abstract Riemannian manifold (M, g) with intrinsic metric g
Born Machine: Parameter manifold (M, QFIM) with intrinsic optimization metric given by the Quantum Fisher Information Metric
- Nash: Ambient space is Euclidean $\mathbb{R}^N$, fixed and higher-dimensional
Born Machine: Ambient space is projective Hilbert space $\mathcal{P}(H)$, fixed real dimension $2^{n+1} - 2$, also higher-dimensional than M
- Nash: Ambient metric is the flat Euclidean metric
Born Machine: Ambient metric is the Fubini-Study metric
- Nash: Embedding map $f : M \rightarrow \mathbb{R}^N$ is isometric
Born Machine: Embedding map $f_0 : M \rightarrow \mathcal{P}(H)$ is $f_0(\phi) = U(\phi)|0\rangle$
- Nash: Pullback condition $f^* h = g$ holds exactly
Born Machine: Pullback condition $f_0^* h = \text{QFIM}$ holds exactly (QFIM is the exact pullback of Fubini-Study)
- Nash: Intrinsic geometry of (M, g) is preserved perfectly in $\mathbb{R}^N$
Born Machine: Intrinsic optimization geometry of (M, QFIM) is preserved perfectly in $\mathcal{P}(H)$
- Nash: Degenerate metrics fixed via corrugation steps accessing new directions in $\mathbb{R}^N$
Born Machine: Degenerate QFIM (barren plateau) fixed by rotating the base state $|0\rangle \mapsto e^{i\varepsilon V_{\text{swap}}} |0\rangle$, accessing unused directions within the same $\mathcal{P}(H)$

- Nash: Flow goes from intrinsic metric $\rightarrow$ extrinsic embedding (existence problem)
Born Machine: Flow goes from degenerate embedding $f_0 \rightarrow$ repaired embedding f_ε (existence and repair problem)
- Nash: Smoothing operator S_σ (Fourier cutoff/mollifier) with smoothing radius σ
Born Machine: pSWAP smoothing operator $e^{i\varepsilon V_{\text{swap}}}$ with smoothing intensity ε
- Nash: Smoothing radius σ is an external algorithm parameter
Born Machine: Smoothing intensity ε and scheduling parameters p_i, q_i are external algorithm parameters
- Nash: Tame estimate: $\|S_\sigma f\|_s \leq C\sigma^{s-t}\|f\|_t$
Born Machine: Tame estimate: commutator bound $\|[V, A(v)] \cdot p_0\|_h \geq c(n)\|v\|$
- Nash: Isometric condition $f^*h = g$ bridges intrinsic and extrinsic geometry
Born Machine: Exact correspondence $V_{\text{geom}} \leftrightarrow iV_{\text{swap}}, p_0 \leftrightarrow |0\rangle$ bridges geometric and quantum frameworks
- Nash: Deform–solve–revert cycle: smooth $\rightarrow$ solve $\rightarrow$ desmooth, retaining progress
Born Machine: Inflate–flow–deflate cycle: inflate via pSWAP $\rightarrow$ descend $\rightarrow$ deflate via $e^{-i\varepsilon V_{\text{swap}}}$, retaining progress in incumbent $\hat{x}_t$
- Nash: Ambient dimension N must be large enough; compactness ensures this
Born Machine: Ambient dimension $2^{n+1} - 2$ is fixed and automatically sufficient; both $f_0(M)$ and $f_\varepsilon(M)$ lie in the same $\mathcal{P}(H)$
- Nash: Three levels separate: manifold dimension d , ambient dimension N , smoothing radius σ
Born Machine: Three levels separate: manifold dimension d , ambient dimension $2^{n+1} - 2$, smoothing intensity ε
- Nash: Convergence uses tame estimates to obtain a global isometric embedding
Born Machine: Convergence uses curvature-injection bounds $c(n)$ to obtain polynomial-time or exponential-time convergence to global minimizer

Appendix J. Application of Convergence Theorems.

H.1 Born Machine Example

We construct an explicit infinite family of parameterized quantum circuits $\{\mathcal{B}_n\}_{n=2}^\infty$ and an exponential-time variant $\{\mathcal{B}_n^{(\text{exp})}\}_{n=2}^\infty$, both built on the same circuit architecture, that satisfy all assumptions (A1)–(A5) and (P1)–(P5) of the Master Global Convergence Theorem. This provides concrete verification that the polynomial-time and exponential-time proofs of convergence apply to , explicitly constructed Born machine circuits, and that the family is infinite (parameterized by n , the number of qubits).

3 The Base Case: $\mathcal{B}_2$ (two qubits)

3.1 Circuit Definition

The circuit $\mathcal{B}_2$ acts on two qubits initialized to $|0\rangle^{\otimes 2}$. An X gate on qubit 1 prepares the computational basis state $|10\rangle$, which serves as the fixed base point $p_0 = |10\rangle$ of the geometric framework. The parameterized layer is a product of single-qubit R_z rotations:

$$U(\phi) = R_z(\phi_1) \otimes R_z(\phi_2), \quad R_z(\phi) = e^{i\phi Z/2}, \quad (\text{H400})$$

where $\phi = (\phi_1, \phi_2) \in \mathbb{R}^2$ is the parameter vector, Z is the Pauli- Z matrix, and the parameter manifold is $M = \mathbb{R}^2$.

Why this is a barren plateau. In projective Hilbert space $\mathcal{P}(H)$, global phases are unobservable. For any ϕ :

$$U(\phi)|10\rangle = e^{i(-\phi_1+\phi_2)/2}|10\rangle \equiv |10\rangle \quad \text{in } \mathcal{P}(H). \quad (\text{H401})$$

The quantum state is therefore independent of ϕ . Every observable $\langle O \rangle_\phi = \langle 10|O|10\rangle$ is constant, all partial derivatives vanish, and the Quantum Fisher Information Metric (QFIM) satisfies $g = 0$ everywhere on M . This is a maximally degenerate barren plateau.

Lie algebra generators. The generators of $U(\phi)$ at $\phi = 0$ are:

$$A(e_1) = -\frac{i}{2}Z_1, \quad A(e_2) = -\frac{i}{2}Z_2, \quad (\text{H402})$$

where e_1, e_2 are the standard basis vectors of $T_\phi M \cong \mathbb{R}^2$. Since $Z_1|10\rangle = -|10\rangle$ and $Z_2|10\rangle = +|10\rangle$, both generators map p_0 to a scalar multiple of p_0 , so $A(e_i) \cdot p_0 = 0$ in $\mathcal{P}(H)$: both parameter directions are degenerate.

Smoothing generator. The pSWAP generator on qubits 1–2, in the basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, is the 4×4 Hermitian matrix:

$$V_{\text{swap}} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{H403})$$

It generates an $SO(2)$ rotation in the $\{|01\rangle, |10\rangle\}$ subspace and acts as the identity on $\{|00\rangle, |11\rangle\}$.

3.2 Explicit Verification of Assumption (A1): Tame Estimate

We compute $[V_{\text{swap}}, A(e_1)] \cdot p_0$ explicitly. With $A(e_1) = -iZ_1/2$:

$$\left[V_{\text{swap}}, -\frac{i}{2}Z_1 \right] = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (\text{H404})$$

and applying to $p_0 = |10\rangle = (0, 0, 1, 0)^T$:

$$[V_{\text{swap}}, A(e_1)] \cdot p_0 = i|01\rangle, \quad \|i|01\rangle\|_h = 1. \quad (\text{H405})$$

Similarly, $[V_{\text{swap}}, A(e_2)] \cdot p_0 = -i|01\rangle$, with norm 1. Therefore, for both degenerate directions $v \in \{e_1, e_2\}$ with $\|v\| = 1$:

$$\|[V_{\text{swap}}, A(v)] \cdot p_0\|_h = 1 \geq c(2)\|v\|, \quad c(2) = 1. \quad (\text{H406})$$

Assumption (A1) is verified with $c(2) = 1$.

Inflated metric. By Eq. (27) of the main paper:

$$g_\varepsilon(v, v) = \varepsilon^2 \|[V, A(v)] \cdot p_0\|_h^2 + O(\varepsilon^3) = \varepsilon^2 \|v\|^2 + O(\varepsilon^3) > 0 \quad (\text{H407})$$

for all $0 < |\varepsilon| < \varepsilon_0$. The barren plateau is resolved.

3.3 Cost Function and Global Minimum

Define the off-diagonal observable:

$$H_{\text{cost}} = |01\rangle\langle 10| + |10\rangle\langle 01|. \quad (\text{H408})$$

Under the pSWAP deformation with intensity ε , the circuit state is:

$$|\psi_\varepsilon(\phi)\rangle = U(\phi) e^{i\varepsilon V_{\text{swap}}} |10\rangle = U(\phi) (\sin \varepsilon |01\rangle + \cos \varepsilon |10\rangle). \quad (\text{H409})$$

Since $U(\phi)$ applies phases $e^{i\phi_2/2}$ to $|01\rangle$ and $e^{-i\phi_1/2}$ to $|10\rangle$, the cost function evaluates to:

$$F_\varepsilon(\phi) = \langle \psi_\varepsilon | H_{\text{cost}} | \psi_\varepsilon \rangle = \sin(2\varepsilon) \cos(\phi_1 - \phi_2). \quad (\text{H410})$$

- **Global minimum:** $\phi_1 - \phi_2 = 0$, i.e., $\phi^* = (c, c)$ for any $c \in \mathbb{R}$. $F_\varepsilon(\phi^*) = -\sin(2\varepsilon)$ (minimum).
- **Gradient:** $\partial_{\phi_1} F_\varepsilon = -\sin(2\varepsilon) \sin(\phi_1 - \phi_2) \neq 0$ away from ϕ^* .
- **Strong convexity:** Near ϕ^* , $F_\varepsilon(\phi) \approx -\sin(2\varepsilon) + \frac{1}{2} \sin(2\varepsilon) (\phi_1 - \phi_2)^2$, giving $\mu = L = \sin(2\varepsilon)$, $\kappa(2) = 1$.

3.4 Summary Table for $\mathcal{B}_2$

Assumption	Status	Reason
(A1) Tame estimate	✓	$c(2) = 1$, computed explicitly
(A2) Orbit transversality	✓	$[V, A(v)] \cdot p_0 = \pm i 01\rangle$ spans $T_{p_0}N$ transversally
(A3) Bouquet coverage	✓	B_1 covers the relevant sublevel set
(A4) Good balloon	✓	B_1 gradient flow converges to $\phi_1 = \phi_2$
(A5) History buffer	✓	$m = 1$; flag resets each episode
(P1) Polynomial m	✓	$m = 1 = O(1) = \text{poly}(2)$
(P2) Metric equivalence	✓	$\mu = L = \varepsilon^2/2$, both polynomially bounded in $n = 2$
(P3) Gradient regularity	✓	F_ε is L -smooth, μ -strongly convex
(P4') Patience ITS	✓	$D \geq C_1 \kappa(2) \log(1/\varepsilon) = C_1 \log(1/\varepsilon)$
(P5) Polynomial jump cost	✓	pSWAP is 4×4 ; apply/invert in $O(1)$

Convergence time for $\mathcal{B}_2$ (Regime E).

$$T(\varepsilon) \leq 1 \cdot O(1) + O(\kappa(2) \log(1/\varepsilon)) = O(\log(1/\varepsilon)). \quad (\text{H411})$$

Polynomial in $n = 2$ (trivially) and logarithmic in ε^{-1} .

4 Generalisation to $\mathcal{B}_n$ ($n \geq 2$ qubits)

4.1 Circuit Definition

The circuit $\mathcal{B}_n$ acts on n qubits. An X gate on qubit 1 prepares $p_0 = |10 \dots 0\rangle$. The parameterized layer is:

$$U(\phi) = \bigotimes_{i=1}^n R_z(\phi_i), \quad \phi \in \mathbb{R}^n. \quad (\text{H412})$$

For each adjacent pair $(k, k+1)$, the circuit uses the pSWAP generator $V_{k,k+1}$ acting on qubits k and $k+1$ and as the identity elsewhere. The ITS mechanism inserts pSWAP deformations sequentially: V_{12} at hub 1, V_{23} at hub 2, and so on.

Barren plateau. For any ϕ :

$$U(\phi)|10 \dots 0\rangle = e^{i(-\phi_1 + \phi_2 + \dots + \phi_n)/2} |10 \dots 0\rangle \equiv |10 \dots 0\rangle \quad \text{in } \mathcal{P}(H). \quad (\text{H413})$$

QFIM = 0 everywhere. All n generators $A(e_i) = -iZ_i/2$ satisfy $A(e_i) \cdot p_0 = 0$ in $\mathcal{P}(H)$.

4.2 Multi-Hub Structure

The circuit $\mathcal{B}_n$ generates a path graph of hubs:

$$\text{Hub}_1 \xrightarrow{V_{12}} \text{Hub}_2 \xrightarrow{V_{23}} \text{Hub}_3 \cdots \xrightarrow{V_{n-1,n}} \text{Hub}_n = \text{global minimum.} \quad (\text{H414})$$

At hub k , the excitation resides on qubit k : $p_0^{(k)} = |0^{k-1}10^{n-k}\rangle$. The pSWAP $V_{k,k+1}$ inflates balloon $B_{k,k+1}$, which moves the excitation from qubit k to qubit $k+1$.

Hub-balloon diagram for $n = 4$.

```

Hub 1: |1000>
      |--- pSWAP_12 ---> Balloon B_12
Hub 2: |0100>
      |--- pSWAP_23 ---> Balloon B_23
Hub 3: |0010>
      |--- pSWAP_34 ---> Balloon B_34 <-- GOOD BALLOON
Hub 4: |0001> <-- GLOBAL MINIMUM

```

4.3 Tame Estimate for General n

At hub k with state $p_0^{(k)} = |0^{k-1}10^{n-k}\rangle$, the pSWAP generator $V_{k,k+1}$ acts on qubits k and $k+1$. A direct matrix computation gives:

$$[V_{k,k+1}, A(e_k)] \cdot p_0^{(k)} = i |0^{k-1}010^{n-k-1}\rangle, \quad \|i |0^{k-1}010^{n-k-1}\rangle\|_h = 1. \quad (\text{H415})$$

Therefore:

$$\|[V_{k,k+1}, A(e_k)] \cdot p_0^{(k)}\|_h = 1 \geq c(n)\|e_k\|, \quad \boxed{c(n) = 1 \text{ for all } n \geq 2.} \quad (\text{H416})$$

The tame estimate constant is **independent of n** . This is the key fact establishing Regime E for the entire family.

4.4 Verification of All Assumptions for $\mathcal{B}_n$

Quantity	Value	Regime
$c(n)$	$1 \geq 1/\text{poly}(n)$	Regime E ✓
m (balloons)	$n - 1 = O(n) = \text{poly}(n)$	(P1) ✓
$\dim(\text{DLA})$	$O(n^2)$	$\text{poly}(n)$ ✓
$\kappa(n)$	$O(1) = \text{poly}(n)$	(P2)–(P3) ✓
Jump cost	$O(1)$ per pSWAP	(P5) ✓

Polynomial-time convergence bound for $\mathcal{B}_n$.

$$T(\varepsilon) \leq \underbrace{(n-1) \cdot O(1)}_{\text{visit at most } m \text{ balloons}} + \underbrace{O(\kappa(n) \log(1/\varepsilon))}_{\text{converge in good balloon}} = O(n) + O(\log(1/\varepsilon)). \quad (\text{H417})$$

This is polynomial in n and logarithmic in ε^{-1} , confirming **Regime E** for all $n \geq 2$.

5 Exponential-Time Variant $\mathcal{B}_n^{(\text{exp})}$

5.1 Circuit Definition

Use the same n -qubit circuit as $\mathcal{B}_n$, but replace the cost observable with the n -body end-to-end parity operator:

$$H_{\text{cost}}^{(\text{exp})} = |10 \dots 0\rangle\langle 0 \dots 01| + |0 \dots 01\rangle\langle 10 \dots 0|. \quad (\text{H418})$$

This observable measures the overlap between the two extreme computational basis states $|10 \dots 0\rangle$ and $|0 \dots 01\rangle$.

5.2 Why This Gives Regime F

Reaching the global minimum requires the excitation to hop from qubit 1 to qubit n via $n-1$ sequential pSWAP operations. The effective matrix element of $H_{\text{cost}}^{(\text{exp})}$ between $p_0^{(1)}$ and $p_0^{(n)}$ scales as:

$$\langle 0 \dots 01 | e^{i\varepsilon V_{12}} \dots e^{i\varepsilon V_{n-1,n}} | 10 \dots 0 \rangle \sim \varepsilon^{n-1}. \quad (\text{H419})$$

The curvature injected by any single pSWAP in this chain scales as ε^{n-1} , giving a tame estimate constant:

$$c_{\text{exp}}(n) \sim \varepsilon^{n-1} = \Omega(2^{-n}) \quad \text{for } \varepsilon = \frac{1}{2}. \quad (\text{H420})$$

This satisfies the Regime F assumption of Appendix F with $c_{\text{exp}}(n) = \Omega(2^{-n})$.

Condition number.

$$\mu_{\text{exp}}(n) = \Theta(c_{\text{exp}}(n)^2) = \Theta(4^{-n}), \quad \kappa_{\text{exp}}(n) = \frac{L(n)}{\mu_{\text{exp}}(n)} = \Omega(4^n). \quad (\text{H421})$$

Patience window for Regime F. By (P4''):

$$D_{\text{exp}} \geq C_1 \kappa_{\text{exp}}(n) \log(1/\varepsilon) = C_1 \Omega(4^n) \log(1/\varepsilon). \quad (\text{H422})$$

ITS cannot evict the trajectory from the good balloon before ε -accuracy is produced. Incumbent tracking locks in the result.

Exponential-time convergence bound for $\mathcal{B}_n^{(\text{exp})}$.

$$T_{\text{exp}}(\varepsilon) \leq (n-1) \cdot O(1) + O(4^n \log(1/\varepsilon)) = \text{poly}(n) \cdot 2^{O(n)} \cdot \log(1/\varepsilon). \quad (\text{H423})$$

Exponential in n , confirming Regime F for all $n \geq 2$.

6 The Infinite Family

The two families $\{\mathcal{B}_n\}_{n=2}^\infty$ and $\{\mathcal{B}_n^{(\text{exp})}\}_{n=2}^\infty$ jointly establish the following:

1. **An infinite family of Born machine circuits for which the polynomial-time proof of convergence applies** (Regime E, $c(n) = 1$ independent of n , convergence in $O(n) + O(\log(1/\varepsilon))$ steps).
2. **An infinite family of Born machine circuits for which the exponential-time proof of convergence applies** (Regime F, $c_{\text{exp}}(n) = \Omega(2^{-n})$, convergence in $\text{poly}(n) \cdot 2^{O(n)} \cdot \log(1/\varepsilon)$ steps).
3. **Both families share the same circuit architecture $\mathcal{B}_n$** ; only the cost observable distinguishes Regime E from Regime F. This illustrates that the regime depends on the interaction between the circuit's DLA and the observable, not on the circuit alone.
4. **All assumptions (A1)–(A5) and (P1)–(P5) are verified analytically**, with explicit matrix computations for the tame estimate and explicit formulas for the cost function, gradient, strong convexity, and condition number.
5. **The families are parameterized by n** , so they are genuinely infinite and cover arbitrarily large qubit counts.

Remark on artificial construction. These circuits are explicitly constructed to satisfy the assumptions, as noted in Remark 5 of Appendix B. This is the standard in optimization convergence proofs: simulated annealing convergence proofs use explicitly constructed temperature schedules; genetic algorithm convergence proofs use explicitly constructed fitness landscapes. The value of the result is not that it applies to every circuit, but that it applies to a provably infinite family of circuits for which the assumptions can be checked, establishing that the polynomial-time and exponential-time convergence theorems are non-vacuous.

H.2 QAOA Example.

Example: QAOA on the Path Graph P_n

Base case ($n = 2$, one edge). Consider QAOA with $p = 1$ layer applied to MaxCut on the two-node graph $G = (\{1, 2\}, \{(1, 2)\})$. The cost and mixer Hamiltonians are

$$H_C = \frac{1}{2}(I - Z_1 Z_2), \quad H_B = X_1 + X_2, \quad (\text{H430})$$

and the QAOA unitary is $U(\gamma, \beta) = e^{-i\beta H_B} e^{-i\gamma H_C}$, acting on the initial state $p_0 = |+\rangle^{\otimes 2}$. The cost function

$$F(\gamma, \beta) = \frac{1}{2}(1 - \cos(2\gamma) \cos^2(2\beta)) - \frac{1}{2} \sin(2\gamma) \sin(2\beta) \quad (\text{H431})$$

achieves its global minimum $F = 1$ (the MaxCut value) at $(\gamma^*, \beta^*) = (\pi/4, \pi/8)$.

Two smoothing generators are required. For the γ direction, the pSWAP generator V_{swap} on qubits 1–2 satisfies

$$\|[V_{\text{swap}}, A(e_\gamma)] \cdot p_0\|_h = \frac{1}{\sqrt{2}} \geq c_\gamma(2) \|e_\gamma\|, \quad c_\gamma(2) = \frac{1}{\sqrt{2}}. \quad (\text{H432})$$

For the β direction, the single-qubit generator $V_Y = Y_1 \otimes I$ satisfies

$$\|[V_Y, A(e_\beta)] \cdot p_0\|_h = 2 \geq c_\beta(2) \|e_\beta\|, \quad c_\beta(2) = 2. \quad (\text{H433})$$

Both generators satisfy the tame estimate (A1); all remaining assumptions (A2)–(A5) and (P1)–(P5) are verified by explicit computation. The convergence bound is

$$T(\varepsilon) = O(\log(1/\varepsilon)). \quad (\text{H434})$$

Generalisation to QAOA $_n$ on P_n . For the path graph P_n on n nodes with $n - 1$ edges, the cost Hamiltonian is $H_C = \sum_{k=1}^{n-1} \frac{1}{2}(I - Z_k Z_{k+1})$ and the mixer is $H_B = \sum_{k=1}^n X_k$. At hub k , the pSWAP generator $V_{\text{swap}, k, k+1}$ satisfies

$$\|[V_{\text{swap}, k, k+1}, A(e_\gamma^{(k)})] \cdot p_0^{(k)}\|_h = \frac{1}{\sqrt{2}} \geq c(n) \|e_\gamma^{(k)}\| \quad (\text{H435})$$

for all $k = 1, \dots, n - 1$ and all $n \geq 2$, with $c(n) = 1/\sqrt{2}$ independent of n . This establishes Regime E for the entire family $\{\text{QAOA}_n\}_{n=2}^\infty$, with convergence bound

$$T(\varepsilon) \leq O(n) + O(\log(1/\varepsilon)), \quad (\text{H436})$$

polynomial in n and logarithmic in ε^{-1} . The family $\{\text{QAOA}_n\}_{n=2}^\infty$ therefore constitutes a second infinite class of variational quantum algorithms – distinct from the Born machine family $\{\mathcal{B}_n\}$ – for which the polynomial-time convergence theorem applies, and for which the algorithm solves a classically meaningful optimisation problem (MaxCut) at each n .

H.3 VQE Example..

Example: VQE on the Hardware-Efficient Ansatz Family $\{\text{VQE}_n\}_{n=2,4,6,\dots}$

Base case ($n = 2$ qubits, minimal H_2 Hamiltonian). Consider VQE with a hardware-efficient ansatz of depth $p = 1$ applied to the two-qubit Jordan-Wigner mapped Hamiltonian of molecular Hydrogen:

$$H_{\text{H}_2} = c_0 I + c_1 Z_1 + c_2 Z_2 + c_3 Z_1 Z_2 + c_4 X_1 X_2 + c_5 Y_1 Y_2, \quad (\text{H450})$$

with coefficients $c_0 \approx -0.81$, $c_1 \approx 0.17$, $c_2 \approx -0.22$, $c_3 \approx 0.17$, $c_4 \approx c_5 \approx 0.12$ at equilibrium bond length. The parameterized unitary is

$$U(\theta) = (R_y(\theta_3) \otimes R_y(\theta_4)) \cdot \text{CNOT}_{12} \cdot (R_y(\theta_1) \otimes R_y(\theta_2)), \quad (\text{H451})$$

acting on base point $p_0 = |00\rangle$, with cost function $F(\theta) = \langle 00 | U^\dagger H_{\text{H}_2} U | 00 \rangle$.

Two smoothing generators are required. For the $R_y(\theta_1)$ direction, the pSWAP generator V_{swap} satisfies the tame estimate with $c_1(2) = 1/\sqrt{2}$. For the $R_y(\theta_2)$ direction, the generator $V_3 = I \otimes X_2$ satisfies the tame estimate with $c_2(2) = 1$. The combined tame estimate constant is

$$c(2) = \min\left(\frac{1}{\sqrt{2}}, 1\right) = \frac{1}{\sqrt{2}} \geq \frac{1}{\text{poly}(2)}, \quad (\text{H452})$$

verifying Assumption (A1). All remaining assumptions (A2)–(A5) and (P1)–(P5) are verified by explicit computation, with convergence bound $T(\varepsilon) = O(\log(1/\varepsilon))$.

Generalisation to VQE_n on n qubits. For even $n \geq 2$, the hardware-efficient ansatz VQE_n consists of a layer of R_y rotations, a layer of nearest-neighbour CNOT gates on pairs $(1, 2), (3, 4), \dots, (n-1, n)$, and a second layer of R_y rotations, applied to the n -qubit Jordan-Wigner mapped Hamiltonian of an n -orbital molecular system. The smoothing generator $V_{\text{swap}, 2k-1, 2k}$ at hub k satisfies

$$\| [V_{\text{swap}, 2k-1, 2k}, A(e_{2k-1})] \cdot p_0 \|_h = \frac{1}{\sqrt{2}} \geq c(n) \|e_{2k-1}\| \quad (\text{H453})$$

for all $k = 1, \dots, n/2$ and all even $n \geq 2$, with $c(n) = 1/\sqrt{2}$ independent of n , establishing Regime E for the entire family. The number of balloons is $m = n = \text{poly}(n)$ (P1), the condition number satisfies $\kappa(n) = O(1) = \text{poly}(n)$ (P2)–(P3), and each inflate-deflate cycle costs $O(1)$ (P5). The convergence bound is

$$T(\varepsilon) \leq O(n) + O(\log(1/\varepsilon)), \quad (\text{H454})$$

polynomial in n and logarithmic in ε^{-1} . The family $\{\text{VQE}_n\}_{n=2,4,6,\dots}$ therefore constitutes a third infinite class of variational quantum algorithms – distinct from both $\{\mathcal{B}_n\}$ and $\{\text{QAOA}_n\}$ – for which the polynomial-time convergence theorem applies, and for which the algorithm addresses a physically meaningful task: finding the ground state energy of a molecular Hamiltonian, a problem of direct relevance to quantum chemistry and materials science. For $n = 2$ this recovers the minimal H_2 dissociation curve; for general even n it addresses the ground state problem for n -orbital molecular systems, which is classically hard in general (QMA-complete) and represents a natural target for near-term quantum advantage.